GEK-73811A

DRIVE SYSTEM INSTRUCTIONS

BUS FED INSTALLATION INSTRUCTIONS

DRIVE SYSTEMS DEPARTMENT
SALEM, VA. 24153

GENERAL ELECTRIC

ISSUE DATE: April 1981

GEK-73811

The information contained herein does not purport to cover all details or variations in equipment nor to provide for every possible contingency to be met in connection with installation, operation, and maintenance. Should further information be desired or should particular problems arise which are not covered sufficiently for the purchaser's purposes, the matter should be referred to General Electric Company, USA.

Nomenclature

Module names and printed wiring boards are underlined for emphasis. Test points are capitalized.

GT DEVICE	STEAM TURBINE DEVICE	DESCRIPTION
26SF	26SF	Static exciter front rectifier bridge temp high
41	41	Generator field breaker
41/PS	41/PS	Generator field breaker position switch
41AC	41AC	Generator exciter supply breaker
41F	53	Generator field flashing contactor
41FX	-	Generator field flashing relay
41FZ	53X	Aux. relay to energize 41F
41S	41S	Generator field suppression aux. relay
41SX	41SX	Generator field suppression aux. relay
41/CS	41/CS	Generator field breaker control switch
43VA	-	VAR-PF controller - selector switch
43VAX	90A	VAR-PF controller aux. relay
49GT	HFTA	Generator field overtemp detector
49GTX	HFTA	Aux. relay to 49GT
52G	52G	Generator breaker
52GX-1	52GX	Generator breaker auxiliary relay
52GX-1A,B	52GX	Aux. relay to 52GX-1
52Z-1	-	Aux. relay to automatic synchronizing
52Z-1X	-	Aux. relay to 52Z-1
55LX	55LX	VAR-PF controller lower relay
55RX	55RX	VAR-PF controller raise relay
59E	59EX	Overexcitation relay
59E-1X	59E	Exciter overvoltage alarm
59E-2X	41T	Exciter overvoltage trip
59E-3X	419X	Aux. relay to 59E-2X
59EZ	59TD	Overexcitation time delay
59VH-1	-	Volts per hertz detector-alarm
59VH-1X	30VZ	Aux. relay to 59VH-1
59VH-2	-	Volts per hertz detector-trip
59VH-2X	41TA	Aux. relay to 59VH-2
60GH	60GH	Generator voltage line voltage relay
60GL	60GL	Generator voltage line voltage relay
64F-TS	64F-TS	Field ground detector test-reset switch
64F2	64F2	Field ground detector timer
64F3	64F3	Field ground detector alarm
83RM	-	Regulator mode (AC/DC) relay
83SR-1	83	Voltage regulator control-transfer relay
83SRX	-	Aux. relay to energize 83SR-1
86G	86G	Generator differential lockout
86T	86T	Transformer differential lockout
90ML	70RL	DC volt reg. setpoint relay-lower
90MR	70RR	DC volt reg. setpoint relay-raise
90RL	90RL	AC volt reg. setpoint relay-raise
90RR	90RR	AC volt reg. setpoint relay-raise
94Z-1	90RB	AC volt reg. setpoint relay-runback
94Z-2	70RB	DC volt reg. setpoint relay-runback

Abbreviations

The following abbreviations are used in this instruction book:

AC, ac	Alternating current
AFFL	Amps, field, full load
AFNL	Amps, field, no load
ARCC	Active and reactive current compensation
CT	Current transformer
D/A	Digital to analog
DC, dc	Direct current
GAC	Generator auxiliary compartment
GEC	Generator excitation compartment
I	Current
IC	Integrated circuit
LDC	Line drop compensator
LED	Light emitting diode
LSB	Least significant bit
MAX	Maximum
MIN	Minimum
MW	Megawatts
PCB	Printed circuit board (same as PWB)
P/B	Pushbutton
PF, pf	Power factor
PPT	Power potential transformer
PS	Power supply
PSS	Power system stabilizer
PT	Potential transformer
PU	Per unit
PWB	Printed wiring board (same as PCB)
RC	Resistor capacitor
RCC	Reactive current compensation
SCR	Silicon controlled rectifier
SVA	Static voltage adjuster
TP, tp	Test point
TVM	Transfer voltmeter
URAL	Under-excited reactive ampere limit
V	Volts
VARS	Volt amperes, reactive
VFFL	Volts, field, full load
VFNS	Volts, field, no load
V/Hz	Volts/Hertz

Test Equipment

The following test equipment is required to correctly set up and calibrate the excitation equipment.

1. One or two digital voltmeters (battery operated or isolated from line).
2. Multimeter (VOM) Simpson or Triplett type.
3. Oscilloscope, 4 MHz dual channel.
4. Clamp-on ammeter.
5. Calculator.

When using the oscilloscope, the scope must be grounded, but do not connect the common clip lead on the probe to the circuit being measured. If there is excessive noise on the signal, or it is distorted, the scope must be isolated (ungrounded) and the clip lead used.

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* WARNING *
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ISOLATION OF TEST EQUIPMENT FROM THE EQUIPMENT UNDER TEST PRESENTS POTENTIAL ELECTRICAL HAZARDS. IF THE TEST EQUIPMENT CANNOT BE GROUNDED TO THE EQUIPMENT UNDER TEST, THE TEST EQUIPMENT'S CASE MUST BE SHIELDED TO PREVENT CONTACT BY PERSONNEL.

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* CAUTION *
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MISOPERATION OF THE EXCITATION SYSTEM WHILE CONNECTED TO THE GENERATOR FIELD CAN CAUSE SERIOUS DAMAGE TO THE EXCITATION SYSTEM AND GENERATOR. DEVIATIONS FROM THE FOLLOWING OPERATIONAL CHECKS AND PROCEDURES SHOULD BE DISCUSSED WITH GENERAL ELECTRIC COMPANY PRIOR TO START-UP.

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* CAUTION *
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TO PREVENT SERIOUS DAMAGE TO THE GENERATOR DURING "BACK FEEDING" THE FOLLOWING PRECAUTIONS MUST BE OBSERVED:

1. DO NOT APPLY VOLTAGES TO THE GENERATOR STATOR (ARMATURE) NOR ROTOR (FIELD) WHILE THE GENERATOR IS AT REST.

2. DO NOT CONNECT NOR APPLY VOLTAGE TO THE EXCITATION SYSTEM.

CHAPTER 2
PRE-EXCITATION CHECKS

Excitation Type:_____

Turoine:_____

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* CAUTION *
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NOT ALL TEST POINTS (2TP ETC.) ARE ISOLATED FROM THE CIRCUIT BY RESISTORS. IF THE TEST POINTS ARE GROUNDED, DAMAGE TO THE AMPLIFIERS MAY RESULT. BE CAREFUL WITH LEADS AND METERS.

Pre-excitation Checks:(Requires approximately 16 hours time)

_____1. Visual inspection; damage, loose wires, etc.

_____2. Wire check per elementaries # _____ (Exc) and # _____ (GCP)

_____3. Check that Regulator Signal CT is located in the proper phase.

_____4. Polarity check all CT's.

_____5. Clean generator panel.

_____6. Clean regulator panels.

_____7. Check forward and reverse resistance of all power diodes and SCR's using a 12 volt battery and 2 ohm resistor.

_____8. Perform mechanical checks of 52G generator breaker.

_____9. Megger stator and field.

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* CAUTION *
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DO NOT HI-POT OR MEGGER ANY CIRCUITS AS THIS MAY RESULT IN DAMAGE TO COMPONENTS. ANY WIRING OR CIRCUITS TO BE HI-POTTED OR MEGGERED MUST FIRST BE DISCONNECTED FROM THE EXCITATION SYSTEM.

	Stator (2500V)		Field (500V)	
	1MIN	10MIN P.I.	1MIN	10MIN P.I.
QA				
QB				
QC				

_____10. Check settings of all protective relays per GCP elementary #_____

_____11. Remove all fuses to 120VAC and 125VDC control circuitry.

_____12. Open PT and open CT shorting switch.

****
* CAUTION *
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CHECK ELEMENTARY FOR EXACT LOCATION AND TERMINAL POINTS.

_____13. Remove all PCB's from slots so that pins are not making contact.

_____14. Energize 125VDC control power.

_____15. Check operation of 83SRX transfer relay (Sh. 09B and 10A). (When completed, place a single pole switch in circuit so pick-up of 83SRX can be controlled at the regulator panel during all checks).

_____16. Check operation of 41F field flashing circuit (Sh. 09A).

_____17. Check operation of 41S suppression circuit (Sh. 09B).

_____18. Check operation of 41AC generator exciter supply breaker (shunt trip, Sh. 09A) or 41 field breaker (Sh.09E).

_____19. Check overexcitation 125VDC control circuitry (Sh. 09A).

_____20. Verify that time delay of 59EZ = 35 seconds (Sh. 09A).

_____21. Check electrical operation of 52G generator breaker.

_____22. Sequence all remaining control circuitry per the elementaries.

_____23. Record all factory potentiometer settings in regulator panel.

(On Interface Module) 90R1 = _____ 90P = _____

90R5 = _____ 90R6 = _____

_____24. Energize 120VAC control power.

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* WARNING *
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WHEN THE FIELD GROUND DETECTOR RELAY IS ENERGIZED, IT APPLIES APPROXIMATELY 100 VOLTS TO THE ENTIRE FIELD CIRCUIT, INCLUDING SEMICONDUCTOR CIRCUITRY AND THE HEAT SINKS, EVEN WHEN THE FIELD IS NOT ENERGIZED.

_____25. Check power supply bus voltages per Table 1, Page 2-7. (Use transformer VRAT for 50 Hz GACS) 41S must be energized.

_____26. Check for proper DC voltages to each gate driver board on the PCM (-18, +36) (41S energized)

_____27. Re-install the PCB's one at a time while monitoring the ± 15 VDC busses for any unusual disturbance caused by a board.

Carefully insert the PWB's.

_____28. Perform the following checks on the exciter regulator:

_____a) AC SVA checks (Page 2-7)

_____b) DC SVA checks (Page 2-8)

_____c) DC Regulator Set Points (Page 2-8)

_____d) Transducer (Page 2-8, 10)

_____e) AC/DC PWB Checks (page 2-10)

_____f) Auto Tracking (Page 2-11)

_____g) Inner Loop Regulator (Page 2-12)

_____h) Field overvoltage checks (Page 2-13)

_____29. Perform firing circuit checks. Open rectifier disconnect switches.

_____a) Synchronizer checks (Page 2-15)

_____b) Reference Phasor checks (Page 2-15)

_____c) Trigger Generator checks (Page 2-17)

_____30. Remove 115 VAC source from the signal conditioner and reconnect the PPT leads. (Sh. 07A) Remove jumper from AC SVA to Inner Loop Regulator.

_____31. Perform the AC Regulator checks (Page 2-14).

_____32. Check 90P, 90R5, 90R6 settings (Page 2-19, 2-20).

_____33. Check operation of the Field Temperature Sensor Board (Page 2-20).

_____34. Check Volts/Hertz trip settings (Page 2-20, 22).

_____35. Check operation of 64F Field Ground Relay (Sh. 10B).

_____36. Check operation of Generator Temperature Indicator, 49G (Gen. Elem).

_____37. Check time delay of Relay 2S = __ Seconds (Gen. Elem).

_____38. Set Load Regulator: (Gen. Elem)

1MW = _____mv

_____a) Spinning reserve = _____mv

_____b) Pre-Selected load = _____mv

_____c) Max Load = _____mv

_____39. Check trips of 86G/HR (Gen. Elem).

_____40. Verify 52G Generator Breaker "Close" circuitry (Gen. Elem).

_____41. Check trips of 52G Generator Breaker (Gen. Elem).

_____42. Check all generator alarms.

_____43. Verify calibration of the DCCT Transducers (Page 2-28, 29).

Check reading of Ammeter on GCP.

_____44. Check that Impedance Compensator Board is set up for DROOP.

Remove from slot so that the pins are not making contact.

_____45. Remove URAL Board from its slot so that the pins are not making contact.

_____46. Perform R and C calculations for URAL board; (Page 2-22, 28).

B = _____

R = _____

C = _____

BJ1 to _____

_____47. Perform Voltage Matching checks (Page 2-28)

_____48. Check operation of Meter Panel (Sh. 10E).

_____49. Verify Hydrogen System is ready.

_____50. Close all Bridge disconnect switches.

_____51. Set 83SRX Test Switch to manual and set 43SR to AUTO.

_____52. Connect test equipment at the regulator panel to be used to monitor the following: a, b, and c are required.

a) Generator Voltage (0-150VAC)

Use DVM and connect from SC-3 to SC-5 (or SC-6)

b) Field Voltage (0-500 VDC)

Connect FVT-10(+), FVT-7 (-); use a Simpson meter.

c) Field Wave Shape (0-10V)

Connect oscilloscope to FVT-4 (+), FVT-5 (-)

or use the differential mode across the field voltage.

d) Field Amps (100mV = 1500A)

Connect FIT -6(+), FIT-8(-)

e) Transfer Voltage (10-0-10VDC)

Connect 5TP(+), 1TP(-) at AC/DC PWB Board

_____53. Close CT Shorting switch and close PT disconnect switch.

_____54. Open $\overline{86G/HR}$ Trip circuit on Sh. 11A to prevent a Turbine trip during off line tests.

This completes the Pre-Excitation checks. Proceed to off line checks to familiarize yourself with the procedures before starting the turbine.

Table 2-1
Power Supply Bus Voltages

	PS4-5	PS5-6	PS7-8	PS9-10	PS10-11
INPUT	+15V Bus	-15V Bus	-24V Bus	36V Bus	-18V Bus
115VAC	+15.0	-15.0	-24.5	+44.0	-21.0
Tolerance	± 0.6	± 0.6	±2.4V	±6.0V	±3.0V

If the measured voltages do not correspond to those given in Table 2-1 within the listed tolerances, investigate the problem before proceeding.

With the Power Supply activated, monitor the ±15 volt power supply busses and install each PC board one at a time. If any board causes unusual voltage fluctuations, do not proceed until the problem has been resolved. The Power Supply should remain connected to the 115VAC source as it will be used in the ensuing tests. All measurements will be made with respect to the common (0 volt) bus unless otherwise directed.

AC SVA Checks

Static Voltage Adjuster (AC SVA) Sh. 05C. Place the AC SVA Counter PWB on the card extender and monitor the SVA output voltage at test point 0 - (+) 10. Exercise the local RAISE pushbutton, the LSB (Least Significant Bit) LED should flash and the voltage increase. It should take approximately two minute to reach the maximum limit of 10 volts. Adjust the GAIN pot, 1P, for a +10 volt output if necessary.

Pot 3P may be adjusted to change the time it takes the SVA counter to run from 0 to +10VDC. This time span should be set to no less than 30 seconds since rapid changes may upset system stability. The nominal setting is 2 minutes from minimum to maximum. For a counting time of 2 minutes, the frequency measured at clock test point on the SVA counter is approximately 36 Hz (28 msec period).

Reinsert the AC SVA Counter PWB into its location in the rack. Exercise the local LOWER pushbutton. The LSB indicator should light and the voltage at 0-(+) 10 should decrease to zero. When power is removed from the SVA, the voltage is reset to zero. Check the operation of 90RL and 90RR by the remote Raise/Lower switch on the Generator Control Panel. (90R4/CS).

Lower the AC SVA to minimum position. Measure voltage at 2TP (T-Ref) of AC SVA Control Board, which is dependent upon setting of 90P. Energize 94Z-1 while monitoring voltage at test point V-OUT on the SVA Control faceplate. The voltage at the test point should track the 90P setting as measured at T-REF. The Raise and Lower LED's may both flash when the output matches. De-energize 94Z-1.

DC SVA Checks(Sh. 06B)

Perform the same checks on the DC SVA as done on the AC SVA (Sh. 8). Use the 90R6/CS on the Generator Panel to control relays 90ML and 90MR. Use 94Z-2 to check the operation of the 90R6 pot. The voltage at V-OUT on the SVA Control PWB should track 90R6 when 94Z-2 is energized. The Raise and Lower LED's may both flash when the output matches.

DC Regulator Set Points (Sh. 06A)

Refer to Note E2 on Sheet A04 of the elementary for set point values for each adjustment.

CONTROL	FUNCTION
2P	Minimum Set Point
1P	Span Adjustment

Check the DC LED on the faceplate of the AC/DC board to ensure the DC regulator is in control of the excitation system. 83SR1 controls the AC or DC Mode.

NOTE: The following setting will be approximate as the regulator is not operating closed loop. The final settings will be performed OFF LINE.

1. MIN Set Point

   Measure the voltage at pins 17, 18 of the AC/DC board and adjust the voltage to zero by 90R5. Measure the voltage at 4TP (DC) and set the MIN to the value on sheet A04, Note E2A, by adjusting 2P(MIN).

2. Span Set Point

   Energize 52GX-1A. Raise 90R6 (or the DC SVA) to maximum limit. Measure the voltage at 4TP and set the span by adjusting 1P(SPAN) on the AC/DC faceplate for the value given on Sh. A04-E2B, de-energize 52GX-1A.

Transducers (Sh. 04B)

The Transducers have been calibrated at the factory for the correct ranges per elementary sheets A04 and 04B. However they must be checked in the field for proper operation. The following checks will ensure that they are operational. Final calibration will be done OFF-LINE.

Voltage Transducer

1. Remove fuses F1FU and F2FU and short the input to the transducer at FVT-7 to FVT-10.

2. Measure the voltage at FVT-4 to FVT-5. Adjust this voltage to 0.00VDC with the BAL pot.

3. Remove the short on the input and connect the 125VDC power to the input. +125VDC to F1FU-2, -125 VDC to F2FU-2. Measure this voltage.

4. Measure the output at FVT-4 to FVT-5. The voltage output should be as follows. Use GAIN to adjust to tolerance.

$$V_{out} = \frac{V_{in}}{V_{cal-in}} \times V_{cal-out} \qquad \text{tolerance } \pm 3\%$$

where V_{cal-in} = calibration input voltage on Sh. A04 Note H

$V_{cal-out}$ = calibration output voltage on Sh. A04 Note H (5VDC)

$$V_{out} = \frac{125}{V_{cal-in}} \times 5 = V_{out\ meas} = ____$$

$V_{out\ meas}$ = _____

5. Remove 125VDC input.

6. If a Field Voltmeter is connected to the transducer, check the current (MA) output by the following:

   A. Short the input again.

   B. Measure the output at FVT-3 to FVT-5. Use a milliammeter, 0-20 milliamperes. The output should be 4.00 ma. Do not adjust at this point if the output is within +.04 ma of this value. If it is greater than + .04 ma, adjust the milliampere output to 4.00 milliamperes with 5P (zero) on the PS/MA board in the transducer.

   C. Remove the short and connect the 125 VDC power to the input again.

   D. The Milliammeter output should be as follows:

$$Ma_{out} = 4 + \frac{V_{in}}{V_{scale}} \times 16 \qquad \text{tolerance } \pm 2\%$$

   Where V_{in} = input voltage to transducer (125 VDC)

   V_{scale} = full scale voltage reading on voltmeter

   $Ma\ out_{cal} = 4 + 125 \times 16 =$ _____

   Do not adjust if the output is within + 2% of the calculated value. (If the output is greater than + 2% of the calculated value, adjust the milliampere output with potentiometer 4P (SPAN) to the calculated value. Repeat steps A-D using both ZERO and SPAN until the values are correct.).

7. Remove the 125VDC and the wire connected to FVT-4. Short this wire to FVT-5.

8. Measure the voltage at pins 22 to pin 24, slot J (Inner Loop Reg). It should be 0.

9. Measure the voltage at 2TP (-VF) on the Inner Loop Regulator PWB. Adjust this voltage to 0.00 volts DC with the CMR potentiometer (2P) on the Inner Loop Regulator.

10. Do not reconnect FVT-4 to prevent back feeding the transducers. This must be done to prevent damage during the next series of tests.

AC/DC PWB (Sh. 06A):

Remove the AC Regulator PWB from the rack assembly. Jumper the AC SVA Counter output, pins 21 and 22 to pins 26 and 27 on the AC/DC board. This simulates an AC Regulator output. Measure the voltage at 4TP on the AC/DC board. (Adjust 90R5 so it is approximately -3.0). Adjust the SVA with the local controls so the voltage at 2TP is the same as that at 4TP. With the voltage of 2TP and 4TP equal, adjust potentiometer 3P (BALANCE) on the AC/DC PWB so that the voltage at 5TP is zero.

Energize 83SR1. The DC LED will go out and the AC LED will light (the Field Overvoltage PWB must be in the rack). The voltage at 3TP will be the opposite polarity of 2TP, and it will be the same as the output of the AC SVA.

At this time, verify the proper operation of the Transfer Volt Meter located at the generator panel. Measure the voltages at 5TP on the AC/DC PWB. The TVM should track 5TP as follows: (See Sheet 06D for calibration).

1. 5TP = 0 Volts ; TVM should be on zero. MA out = 12.0
   AC (2TP) = DC (4TP)

2. 5TP = +10 Volts ; TVM should deflect to the right of zero.
   AC(2TP) DC (4TP) MA out = 20

3. 5TP = -10 Volts ; TVM should deflect to the left of zero.
   AC(2TP) DC (4TP) MA out = 4

If TVM does not work properly, check for mis-wiring or mis-calibration of the Volt-MA Converter board. Refer below for calibration information. Check the operation of 83RM (Sh. 10D elementary) and lights on GCP. De-energize 83SR1.

Volts/MA Converter Calibration (Sh. 06D)

1. Monitor the voltage at 5TP (TR UNBAL) on the AC/DC PWB. Adjust the AC SVA output simulating the AC Regulator output until the voltage at 5TP is 0.0 VDC.

2. Adjust the VOLTS/MA Converter #3 Zero pot to give a 12.0 milliampere output. Use the SPAN potentiometer to get close.

3. Measure the voltage at 5TP (TR UNBAL) on the AC/DC PWB. Adjust this voltage to +10.0VDC using the AC SVA.

4. Adjust the VOLTS/MA Converter #3 SPAN potentiometer to give a 20 milliampere output.

5. Measure the voltage at 5TP on the AC/DC PWB. Adjust this value to -10VDC by the AC SVA output simulating the AC Regulator. The current output should be 4.0 milliamperes.

6. Repeat steps 3-5 until no further adjustment is required.

7. Note: At ± 10VDC (the TR UNBAL signal) the milliampere output need not be exact because the most significant point is when TR UNBAL is 0.0 VDC (12 Milliamperes).

8. Repeat steps for other signals (FVT, FIT, etc).

Automatic Tracking Adjustment (Sheet 06C)

1. Place the Transfer and Tracking PWB on the extender in its location in the rack 1M4B.

2. Monitor the clock signal at 8TP with an oscilloscope. Adjust the period to 3.0 milliseconds using the TIME potentiometer 2P.

3. Close relay 83SR1. The AC LED on the AC/DC PWB should be lit. Energize relay 52GX-1A.

4. Measure the voltage at 7TP. Adjust this value to 6.3 (± .05)VDC with the DEADBAND potentiometer 7P. This may be adjusted later.

5. Monitor the voltage at 5TP on the AC/DC PWB (marked TRUNB on the faceplate). This will be called the TRUNB signal.

6. Use the RAISE and LOWER pushbuttons on the AC SVA Control PWB to adjust the TRUNB signal to a positive value.

7. Measure the voltage at 3TP on the Transfer and Tracking PWB. It should be -13 (±1.5)VDC. Measure the voltage at 4TP. It should be +10 (±0.5) VDC. The RAISE LED will be lit.

8. Verify that after a time delay of 30 seconds, the RAISE LED will flash and the output of the DC SVA Counter PWB at the test point marked 0-(+) 10 will increase. Make sure 94Z-2 is deenergized.

9. Lower the AC SVA setting with the LOWER P/Bs until the TRUNB signal becomes negative.

10. Measure the voltage at 3TP. It should be +10 (±0.5) VDC. Measure the voltage at 4TP. It should be -13 (±1.5) VDC. The LOWER LED will be lit.

11. Verify that after a time delay of 30 seconds, the LOWER LED will flash and the output of the DC SVA Counter PWB at the test point marked 0-(+)10 will decrease.

12. Remove the Transfer and Tracking PWB from the extender and place it in its location 1M4B.

13. When the Transfer and Tracking PWB is operating correctly, it may over shoot the AC setpoint slightly and then go the opposite way and again overshoot. This is OK. But to minimize the overshoot, adjust DEADBAND. Turn 1P CCW to decrease deadband. The Transfer and Tracking should be adjusted so that the TRUNB signal is within 1.0 volts of zero on the TVM.

14. When checks are complete, remove the jumper from the AC SVA Counter output to Pins 26 and 27 of the AC/DC PWB. Dropout 52GX-1A.

Inner Loop Regulator (Sh. 06A)

The following controls will be adjusted:

CONTROL	FUNCTION
5P	Bias(Limit)
1P	Field Voltage Signal Gain
6P	Inner Loop Gain

Remove the Field Overvoltage PWB. Ensure that the field voltage signal from the Transducer is removed on Pin 22. (Wire removed from FVT-4). Energize 41S.

Jumper the AC SVA Counter output, (pin 21) to pin 22 on the Inner Loop Regulator Board. Remove only the one wire from the transducer, as a ground return path is required. The AC SVA output is used as a simulated field voltage signal. Using the calibration data on A04, E1A and H determine what Field Voltage transducer output will produce a -5.0VDC -VF signal.

$$V_{out} = \frac{\text{Inner Loop field volts calibration value}}{\text{Transducer field volts calibration value}} \times \text{transducer output calibration value}$$

V_{out} = _____

Where V_{out} is the value of transducer output voltage that produces a -5.0 VDC V_f signal.

and "Inner Loop Field Volts Calibration Value" is the field voltage value in note E1A that gives -5.0 VDC at 2TP.

and "Transducer Field Volts Calibration Value" is the field voltage value in Note H for the DCPT

and "Transducer Output Calibration Value" is the output voltage value given in Note H.

Using the local AC SVA controls, set the voltage at pins 22 and 23 to the value calculated. Monitor the voltage at 2TP on the Inner Loop Regulator board and adjust the GAIN, 1P, for -5.0VDC. With -5.0 volts at 2TP, the INNER LOOP GAIN can now be adjusted using the following procedure. Measure the voltage at 3TP and 4TP. By adjusting the DC SVA or 90R5, make sure that 3TP is within ± 7.0 VDC of zero. Adjust the LOOP GAIN, 6P such that the magnitude of V_{3TP} gives the

$\overline{V_{4TP}}$

$$\left| \frac{V3TP}{V4TP} \right| \times 10 = \text{LOOP GAIN} _____$$

The BIAS (LIMIT) Pot 5P, adjusts the maximum value of the -ER signal (ER at 6TP). With 0 volts at 3TP, 6TP should be between -7 and -9 VDC. Do not adjust potentiometer 5P at this time as it will be checked on page 2-17. Continue to use the AC SVA output to simulate the field voltage signal.

Field Overvoltage PWB (Sh. 10A)

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* CAUTION *
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OPERATION OF THE FIELD OVERVOLTAGE BOARD WILL CAUSE 59E-1X AND 59E-2X TO ENERGIZE THESE DEVICES WILL ACTIVATE THE ANNUNCIATOR AND CAUSE A TRIP AT 86G/HR.

Place the Field Overvoltage PWB on the extender and insert it into the rack. The pulse train periods, test points, and controls are given on sheet A04, Note F of the elementary.

1. Measure the voltage at 2TP on the Inner Loop Regulator PWB. Adjust this voltage to the OFF LINE set point given on Sheet A04 F4 with the AC SVA.

2. Using an oscilloscope, monitor test point 2TP on the Field Overvoltage PWB.

3. Rotate potentiometer 1P CW until the voltage at 2TP is -sat.

4. Rotate potentiometer 1P CCW until 2TP just goes to +sat. At this point the time delay will start.

5. Decrease the AC SVA until the 2TP voltage goes negative again.

6. Using an oscilloscope, check the clock pulse period at 9TP. It should be 4 milliseconds. Adjust 3P until this condition is met.

7. Increase the AC SVA until 2TP goes to +sat. Start timing at this point. 10 seconds later, the ALARM LED and the BKR TRP LED should flash.

8. Decrease the AC SVA below the set point and the LEDs should go out.

9. Repeat the above test with 83SR1 energized. The LEDs must be reset by the reset pushbutton.

10. Energize 52GX-1A and 83SR1.

11. Measure the voltage at 2TP on the Inner Loop Regulator PWB. Adjust this voltage to the ON LINE setpoint given on sheet A04 F2.

12. Using an oscilloscope, monitor test point 4TP on the Field Overvoltage PWB.

13. Rotate potentiometer 2P CW until the voltage at 4TP is -sat.

14. Rotate potentiometer 2P CCW until the voltage at 4TP just goes to +sat. At this point, the time delay will start.

15. Decrease the AC SVA until the 4TP voltage goes negative again.

16. Using an oscilloscope, check the clock pulse period at 7TP. It should be 15 milliseconds. Adjust 4P to meet this condition. Check the clock pulse period at 8TP. It should be 10 milliseconds. Adjust 5P to meet this condition.

17. Increase the AC SVA until 4TP goes to zero. Start timing at this joint. 15 seconds later the ALARM LED should light, 59E-1X should energize and the TCP annunciator should light. The AC LED on the AC/DC PWB should go out and the DC LED should go on. After an additional 10 seconds, the BKR TRP LED should go on and 59E-2X should energize and 41AC should trip. The voltage at 2TP on the Inner Loop Regulator should be at the value given on A04-F2.

18. Decrease the AC SVA below the set point and reset the LEDs with the pushbutton. The LEDs must always be reset when 83SR1 is energized.

19. Remove the PWB from the extender and replace in the rack. De-energize 52GX-1A and 83SR1.

20. Do not remove the jumper simulating field voltage at this time.

The secondary relays 59E and 59EZ are preset at the factory at the voltage value on sheet A04 and 35 seconds respectively. Refer to the PJV instructions for adjustment procedures. Verify the operation of 59EZ.

Synchronizer (Sheet 07A)

Disconnect the PPT signal from the Signal Conditioner module at terminals SC8 and SC10 or SC12. Connect a 115 VAC source to SC8, SC10 and check SC12. Also check the signals on the Synchronizer board. Put the oscilloscope on test point 5TP (-E SYNCH) the voltage should be a square wave form clipped at about ± 10V. Check the Q_A and I_A signals from the synchronizer with the scope. Q_A should be 6.36 VRMS. If necessary, adjust Q_A magnitude with potentiometer 3P. The I_A magnitude should be close to Q_A. These signals are necessary to obtain the correct vector relationship from the reference phasor. Replace card if they cannot be obtained.

Reference Phasor (Sheet 07A) See Figure 2-1.

1. Place the Reference Phasor on the extender.

2. Use the oscilloscope to view Q_A on CH 1. Set the time sweep so that 1/2 complete cycle is 9 divisions. Trigger on Line Synch. Use CH 2 to view Q_B. It should lag Q_A by 120° (6 divisions ± 0.1 division). Adjust the lag phase by potentiometer 5P. Measure the amplitude with a DVM. Set the amplitude of Q_B to exactly the same value as Q_A (within 0.01 VAC) with potentiometer 6P (Do not readjust potentiometer 5P).

3. Connect CH 2 to the QC signal, 4TP. QC should lead Q_A by 120° (6 divisions ± 0.1 division). Adjust the lead phase angle by potentiometer 3P. Measure the amplitude with a DVM. Set the amplitude of QC to the exact same value as Q_A (within 0.01 VAC RMS) with potentiometer 4P. (Do not readjust potentiometer 3P).

4. Connect CH 1 to the IA signal on 4TP of the Synchronizer PWB. Adjust the scope trigger so that 9 divisions equals 1/2 cycle.

5. Connect CH 2 to the $-I_B$ (3TP on the Reference Phasor) signal. $-I_B$ must lead IA by 60° (3 divisions ± 0.1 divisions). Adjust the phase angle by potentiometer 1P.

6. Connect CH 2 to the -IC (2TP on the Reference Phasor) signal. -IC must lag IA by 60° (3 divisions ± .1). Adjust the phase angle by potentiometer 2P.

7. The -IA, -IB, -IC signals must be sinusoids with the phase relationships shown on Figure 2-1. Since the magnitudes of the signals are not used in the trigger generator (only the zero crossing is used), the magnitude of each signal is not critical.

8. Replace the Reference Phasor PWB into the rack.

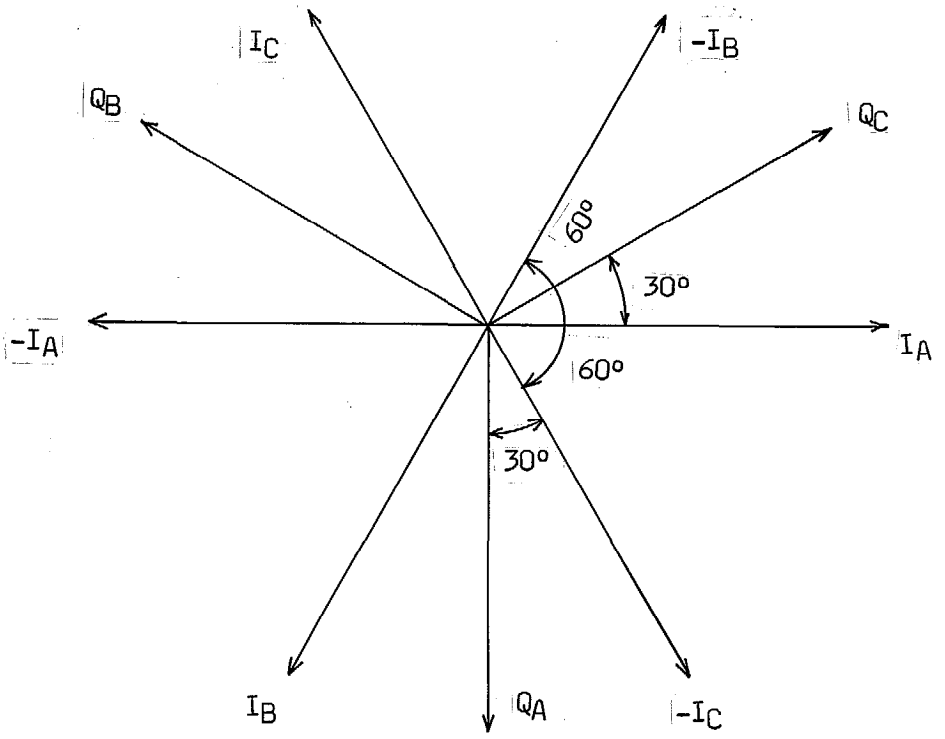


Figure 2-1
Q and I Vectors in Reference Circuit

Signals generated by Synchronizer Board and Reference Phasor Board.

NOTE

If Q signals are not equal in amplitude, you can only adjust the following by: (Amplitude Adjustment)

1) Q_A; adjust potentiometer 3P on Synchronizer Board.

2) Q_B; adjust potentiometer 6P on Reference Phasor Board

3) Q_C; adjust potentiometer 4P on Reference Phasor Board

(No adjustment for I_A, $-I_A$, $-I_C$)

Trigger Generator and Gate Drivers (Sh. 07B)

Check to ensure the regulator is in the DC mode and the AC SVA at full lower. Energize 41S. Monitor the pulse train output from the Trigger Generator boards with the scope at 5TP (OUT). The signal should be similar to Figure 2-2. Use the AC SVA to adjust the simulated field voltage signal until the trigger outputs appear. (You should have jumper connected from AC SVA Counter board, pins 21 and 22 to the Inner Loop Board, pin 22. Note that when the $-V_F$ signal is raised, the overvoltage setpoint may be exceeded and 59E-1X and 59E-2X may pick up. Remove the relays if desired.

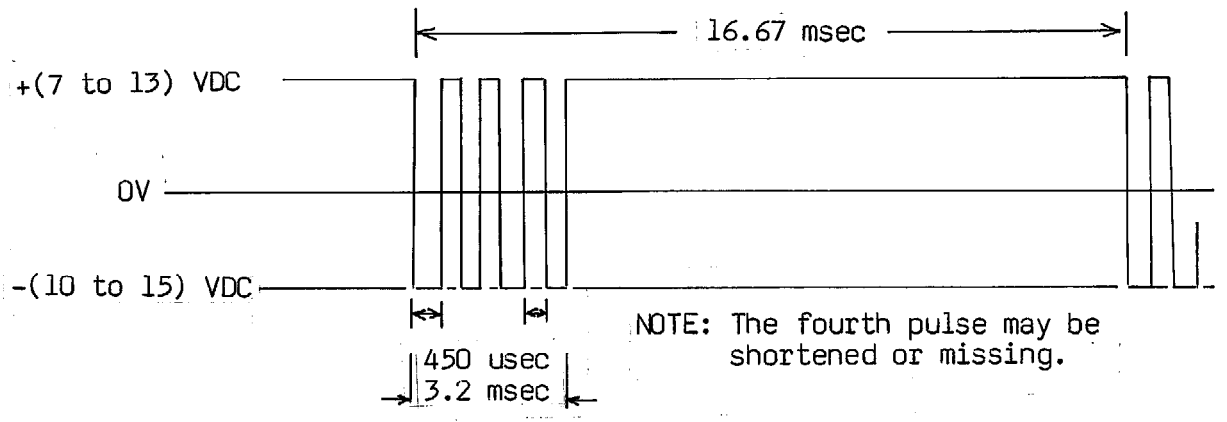


Figure 2-2
Trigger Generator Output (5TP)

Check Trigger Generators A, B and C; each should have the same output displaced by 120 degrees (5.56 milliseconds at 60 Hz). Set the oscilloscope to line synchronized and raise the SVA. The pulse train should move to the right indicating an increasing firing angle and decreasing excitation.

Raise the SVA until the trigger pulses have stopped moving to the right. This point is the phase back limit (if the pulses have disappeared, turn 5P to bring the pulses back). Adjust the scope to begin the sweep at the earliest firing point (trigger pulses fully to left). Adjust potentiometer 5P so that the pulses will only travel 160° (7.4 msec) when the SVA is fully raised.

Place CH1 on Trigger Generator A output (5TP). Use CH1 to check each bridge Gate Driver. All phase A SCR's Gate Drivers must have phase A Trigger Generator outputs. Phase B or C will have CH2 displays that are shifted 120°. Refer to the elementary for pin numbers on the Gate Driver cards.

Repeat the above for each phase.

Check the signals at the output of the Gate Drivers (SCR Gates) on all bridges with an ungrounded scope.

****
* WARNING *
****

ISOLATION OF TEST EQUIPMENT FROM THE EQUIPMENT UNDER TEST PRESENTS POTENTIAL ELECTRICAL HAZARDS. IF THE TEST EQUIPMENT CANNOT BE GROUNDED TO THE EQUIPMENT UNDER TEST, THE TEST EQUIPMENT'S CASE MUST BE SHIELDED TO PREVENT CONTACT BY PERSONNEL.

The output should be similar to that in Figure 2, but reduced in magnitude (3 Volts peak to peak) and inverted.

Disconnect the 115 volt supply to the Signal Conditioner and reconnect the PPT input taking care to make the correct connections to SC8, SC10 and SC12 according to the elementary. Remove the jumper from the SVA to pins 22 and 23 on the Inner Loop Regulator PWB and reconnect the Field Voltage Transducer signal at the Interface module. De-energize 41S.

AC Regulator (Sh. 05B)

The following controls will be adjusted. Note: Remove optional PWBs such as URAL, V Hz and Impedance Compensator. De-energize 52GX-1A.

CONTROL	FUNCTION
2P	Minimum set point
1P	Voltage adjust span
3P	AC REgulator board gain
5P	Lag time constant
4P	Lead time constant

The gain, lead and lag for each specific unit will be found on sheet A04, Note 3 of the elementary. All preliminary adjustments should be made to agree with those specified. Remove the AC Voltage Sensor board from the rack and insert the AC Regulator board. Jumper the output of 90R1 to pins 21 and 22 of the AC Regulator Board. This simulates the -VT signal. Set the voltage at pins 21 and 22 to -4.5 volts using 90R1. Run the AC SVA to 0 volts by the local pushbutton and monitor the voltage at 4TP on the AC Regulator faceplate.

Adjust the MIN potentiometer, 2P such that 4TP is at 0 VDC.

Set the voltage at pins 21 and 22 on the AC Regulator PWB to -5.5 VDC using 90R1 and run the AC SVA to the full raise position. Adjust the SPAN potentiometer, 1P such that 4TP is at 0 VDC.

Set 90R1 for -5.0 volts at pins 21 and 22; the SVA should remain in the full raise position. The output at 2TP on the AC/DC board should be -10 $\pm$ 1 D .

Lower the AC SVA until 4TP is close to $\pm$ 1 VDC.

Repeat the above for each phase.

Check the signals at the output of the Gate Drivers (SCR Gates) on all bridges with an ungrounded scope.

****
* WARNING *
****

ISOLATION OF TEST EQUIPMENT FROM THE EQUIPMENT UNDER TEST PRESENTS POTENTIAL ELECTRICAL HAZARDS. IF THE TEST EQUIPMENT CANNOT BE GROUNDED TO THE EQUIPMENT UNDER TEST, THE TEST EQUIPMENT'S CASE MUST BE SHIELDED TO PREVENT CONTACT BY PERSONNEL.

The output should be similar to that in Figure 2, but reduced in magnitude (3 Volts peak to peak) and inverted.

Disconnect the 115 volt supply to the Signal Conditioner and reconnect the PPT input taking care to make the correct connections to SC8, SC10 and SC12 according to the elementary. Remove the jumper from the SVA to pins 22 and 23 on the Inner Loop Regulator PWB and reconnect the Field Voltage Transducer signal at the Interface module. De-energize 41S.

AC Regulator (Sh. 05B)

The following controls will be adjusted. Note: Remove optional PWBs such as URAL, V Hz and Impedance Compensator. De-energize 52GX-1A.

CONTROL	FUNCTION
2P	Minimum set point
1P	Voltage adjust span
3P	AC REgulator board gain
5P	Lag time constant
4P	Lead time constant

The gain, lead and lag for each specific unit will be found on sheet A04, Note 3 of the elementary. All preliminary adjustments should be made to agree with those specified. Remove the AC Voltage Sensor board from the rack and insert the AC Regulator board. Jumper the output of 90R1 to pins 21 and 22 of the AC Regulator Board. This simulates the -VT signal. Set the voltage at pins 21 and 22 to -4.5 volts using 90R1. Run the AC SVA to 0 volts by the local pushbutton and monitor the voltage at 4TP on the AC Regulator faceplate.

Adjust the MIN potentiometer, 2P such that 4TP is at 0 VDC.

Set the voltage at pins 21 and 22 on the AC Regulator PWB to -5.5 VDC using 90R1 and run the AC SVA to the full raise position. Adjust the SPAN potentiometer, 1P such that 4TP is at 0 VDC.

Set 90R1 for -5.0 volts at pins 21 and 22; the SVA should remain in the full raise position. The output at 2TP on the AC/DC board should be -10 $\pm$ 1 D .

Lower the AC SVA until 4TP is close to $\pm$ 1 VDC.

Measure and record the magnitudes of the voltages at 4TP, 5TP, 6TP, 7TP and 8TP and calculate the GAIN, LAG and LEAD.

$$\text{GAIN, } K_3 = \text{V5TP/V4TP}$$

$$\text{LAG, } K_5 = \text{V8TP/V7TP}$$

$$\text{LEAD, } K_4 = \text{V6TP/V5TP}$$

Use the AC SVA to make sure no test points are at saturation ($\pm$ 9V).

If the calculated K_3, K_4, and K_5 do not agree with the specification on sheet A04 of the elementary, use the specified K_3, K_4 and K_5 to calculate the test point voltages.

1. GAIN, K_3: measure the magnitude of voltage at 4TP, and adjust 3P, GAIN such that:
$$V_{5TP} = V_{4TP} \times K_3$$
2. LAG: Measure the magnitude of the voltage at 7TP and adjust 5TP, LAG, such that:
$$V_{8TP} = V_{7TP} \times K_5$$
3. LEAD: Measure the magnitude of the voltage at 5TP and adjust 4P, LEAD, such that:
$$V_{6TP} = V_{5TP} \times K_4$$

Exercise the SVA to the full lower position and check the regulator output at 2TP on the AC/DC PWB. The voltage should go to +8.5 $\pm$ 2.5 VDC. Readjust 90R1 to +1.0 VDC, reconnect 90R1 as shown in the elementary.

Reinsert the AC Voltage Sensor board.

90P Setting:

Measure the voltage at 2TP (T-REF) of the AC SVA Control Board and set voltage per the value on Sh. A04 - D3G by adjusting potentiometer 90P. Lock 90P.

90R5, Off-Line Set Point

Adjust 90R5 so the voltage at 4TP on the AC/DC board is the value shown on sheet A04, note E2C. This value is approximately equivalent to VFNL at 25°C. Potentiometer 90R5 should be locked at this setting until the generator is running off-line at no load. If required, 90R5 will then be adjusted to hold rated terminal voltage in the DC regulator mode.

90R6, On-Line Set Point

Energize 52GX-1A. Measure the voltage at 4TP on the AC/DC PWB. Adjust this voltage to the value given on sheet A04, note E2D. Adjust with 90R6. Lock knob.

NOTE: For excitation systems with the DC SVA option, adjust 4TP with the DC SVA. Measure V_{out} on the DC SVA Control PWB and set the voltage at TREF (same PWB) with 90R6 to the same value. Lock knob.

Field Temperature Indicator Checks (Sh. 04C)

Remove the DCPT input to the regulator at FVT-5 (as in the Inner Loop check).

Remove the DCCT input to the Temperature Sensor Board at Pin 29.
Jumper Pin 27 to Pin 28 on the Temperature Sensor PWB and tie to 0 volt bus (COM).
Measure voltage at 5TP. Adjust to 0 volts with 12P. Remove jumper.
Jumper Pin 29 to Pin 30 and tie to 0 volt bus (COM).
Jumper 4TP to 6TP. Adjust 13P for 0 volts at 4TP.
Remove jumpers.

Perform calibration checks of Temperature Sensor Board per information provided in elementary Sh. 04C.
To simulate VF signal, use the AC SVA. Jumper wire from AC SVA Control Board, pins 21 and 22 and connect to Temperature Sensor Board, Pin 27. To simulate IF signal, use DC SVA jumper wire from DC SVA Control Board, Pins 21 and 22 and connect to the Temperature Sensor Board, Pin 29.

Use the values on sheet 04C, steps 1-10. The numbers in parenthesis are the values of the AC SVA and DC SVA outputs respectively. Use an oscilloscope for the indication at 8TP for step #1.

When calibration is complete, restore all wiring to original connections per elementaries.

V/Hz Trip Setting (Sh.10C)

To perform this test, a variac is required. Use either a 3 phase or single phase variac.

Single Phase Sensing

Connect the variac to the Signal Conditioner SC-3 and SC-6 (Disconnect PT feed). Only single phase is required. Adjust the variac for rated PT output. Measure the secondary voltage of the signal conditioner. Adjust the variac until the secondary voltage REC 2 to SC18 is at 18 volts AC

3 Phase Sensing

Connect the 3 phase variac to the Signal Conditioner input SC-3, SC-4, SC-5 (Disconnect PT feed).

OR

Connect the 1 phase variac to transformer 3T, SC-5 to TB-5. This places a single phase voltage across the primary (1-3) of 3T. (Disconnect PT feed).

Adjust the variac until the secondary voltage SC-22 to SC-16 is at 17VAC.

1. Monitor the voltage at 2TP on the V/Hz Input PWB. Adjust this value to -5.00 (± 0.01)VDC with the GAIN potentiometer 2P.

2. Monitor the voltage at 3TP (LO REF) on the V/Hz Input PWB. Adjust this value to 11.00 (± 0.02) VDC using potentiometer 3P.

3. Monitor the voltage at 5TP (HI REF) on the V/Hz Input PWB. Adjust this value to 12.00 (± 0.02) VDC using potentiometer 6P.

4. Slowly raise the output of the variac until the voltage at 2TP reaches -5.90 (± 0.01)VDC.

5. Measure the voltage at 4TP on the V/Hz Input PWB. Adjust this value to 10.00 (± 0.01) VDC using potentiometer 4P.

6. Monitor the square wave at 6TP with an oscilloscope. Adjust the period (1 cycle) to 4.5 milliseconds using potentiometer 5P.

7. Lower the output of the PT sensing variac until the voltage at 2TP reaches -5.00 (± 0.01) VDC. If the TRIP LED is flashing, operate the Reset pushbutton.

8. Close the TEST SWITCH. Monitor the square wave at 6TP with an oscilloscope. Adjust the period of this signal to 4.5 milliseconds using TEST potentiometer 1P. Turn the TEST SWITCH off and RESET.

9. Close TEST SWITCH again and check that the TRIP LED will begin to flash 45 seconds later. The TRIP TIMING LED should begin to flash shortly after the TEST SWITCH is turned on. The ALARM relay 59VH-1X should be energized. Turn off TEST SWITCH and RESET.

10. Monitor the voltage at 2TP on the V/Hz Input PWB. Close the TEST SWITCH and adjust the voltage at 2TP to -5.92 (± 0.01) using the TEST potentiometer 1P.

11. Monitor the square wave at 6TP. Reduce the setting of the HI REF potentiometer 6P just until the frequency of the square wave switches to a higher value. Verify that the voltage at 5TP is 11.80 (±0.02) VDC. If not, repeat steps 10 and 11.

12. Adjust the period of the new frequency at TP6 to 0.2 milliseconds using potentiometer 7P.

13. Open the TEST SWITCH and RESET. Verify the values in the following table by adjusting the output of the PT sensing variac to obtain the indicated voltages at 2TP.

V (2TP)	Period at 6TP
-5.90 (±0.01)	4.5 milliseconds
-5.80 (±0.01)	6.0 milliseconds
-5.70 (±0.01)	9.0 milliseconds
-5.60 (±0.01)	18.0 milliseconds

14. If these values are not met, repeat steps 6 - 12.

15. Decrease the PT sensing variac until the voltage at 2TP reaches -5.00 (± 0.01) VDC. Raise it again until 2TP reaches -5.90 (± 0.01)VDC. Verify that after a 45 second time delay, the TRIP LED on the Output PWB will begin to flash and the TRIP relay 59VH-2X on the interface module will be energized. Verify the operation of the contacts shown on sheet 09A.

16. Remove the variac and reconnect PT feed.

URAL Calculations (Sh. 05B) (Refer to GEK-83767 for URAL 11)

URAL Setting (Partial). The customer should specify the desired shape of the limit curve, including the circle center, the 0 MW axis crossing, and the 0 MVAR axis crossing. If no limit characteristic curve is available from the customer at this time, set the limit as follows:

A. See the D235T8 capability curve (Figure 2-4) for an example.

B. Assume the under-excited axis crossing at 60% of the generator VAR capability at 0 PF, under-excited (this is Y).

C. Assume the unity PF axis crossing at 1.1 PU MW (this is X).

D. Calculate the circle center on the vertical axis (B = 90°) using equation 2, below).

Knowing the desired curve, the distance of the circle center from the origin of the machine reactive capability curve and the radius of the circle can be determined in PU.

Figure 2-3 shows the method of construction of the characteristic curve on a typical generator capability curve. It is shown in actual values, but the equations that follow can be used with either PU or actual volt-ampere quantities, whichever is most convenient. It is necessary to convert to PU the values for circle center and circle radius before calculating the voltages that determine those parameters. Those voltages will be set later when the generator is on-line.

The method for determining the values of C and R can best be shown by an example. Refer to Figures 2-3 and 2-4, and Table 2-2 for SIN/COS values.

The equations that apply are:

$$\textcircled{1}\quad C = \frac{X^2 - Y^2}{2(X\mathrm{Cos}B + Y\mathrm{Sin}B)} \qquad R = \frac{\sqrt{(X^2 + Y^2)^2 + (X^2 + Y^2)\ \mathrm{Sin}\ (2B)}}{2(X\mathrm{Cos}B + Y\mathrm{Sin}B)}$$

NOTE: For 3 Phase Sensing Only:

If the circle center is not on the ordinate (vertical axis), it is necessary to accurately construct the limit curve to get a value of the angle B. Once this has been done, the values of C and R can be calculated, so the above equation is necessary.

Table 2-2
SIN/COS TABLE

ANGLE	SIN	COS	ANGLE	SIN	COS
0	.000	1.000	45	.707	.707
1	.017	1.000	46	.719	.695
2	.035	.999	47	.731	.682
3	.052	.999	46	.743	.669
4	.070	.998	49	.755	.656
5	.087	.996	50	.766	.643
6	.105	.995	51	.777	.629
7	.122	.993	52	.788	.616
8	.139	.990	53	.799	.602
9	.156	.988	54	.809	.588
10	.174	.985	55	.819	.574
11	.191	.982	56	.829	.559
12	.208	.978	57	.839	.545
13	.225	.974	58	.848	.530
14	.242	.970	59	.857	.515
15	.259	.966	60	.866	.500
16	.276	.961	61	.875	.485
17	.292	.956	62	.883	.470
18	.309	.951	63	.891	.454
19	.326	.946	64	.988	.439
20	.342	.940	65	.906	.423
21	.358	.934	66	.913	.407
22	.375	.927	67	.920	.391
23	.391	.921	68	.927	.375
24	.407	.914	69	.933	.359
25	.423	.906	70	.940	.342
26	.438	.899	71	.945	.326
27	.454	.891	72	.951	.309
28	.469	.883	73	.956	.293
29	.485	.875	74	.961	.276
30	.500	.866	75	.966	.259
31	.515	.857	76	.970	.242
32	.530	.848	77	.94	.225
33	.545	.839	78	.978	.208
34	.559	.829	79	.982	.191
35	.573	.819	80	.985	.174
36	.588	.809	81	.988	.157
37	.602	.799	82	.990	.139
38	.616	.788	83	.993	.122
39	.629	.777	84	.994	.105
40	.643	.766	85	.996	.087
41	.656	.755	86	.998	.070
42	.699	.43	87	.999	.053
43	.682	.731	88	.999	.035
44	.695	.719	89	1.000	.018
45	.707	.707	90	1.000	.000

FOR NEG ANGLES SIN = NEG, COS = POS

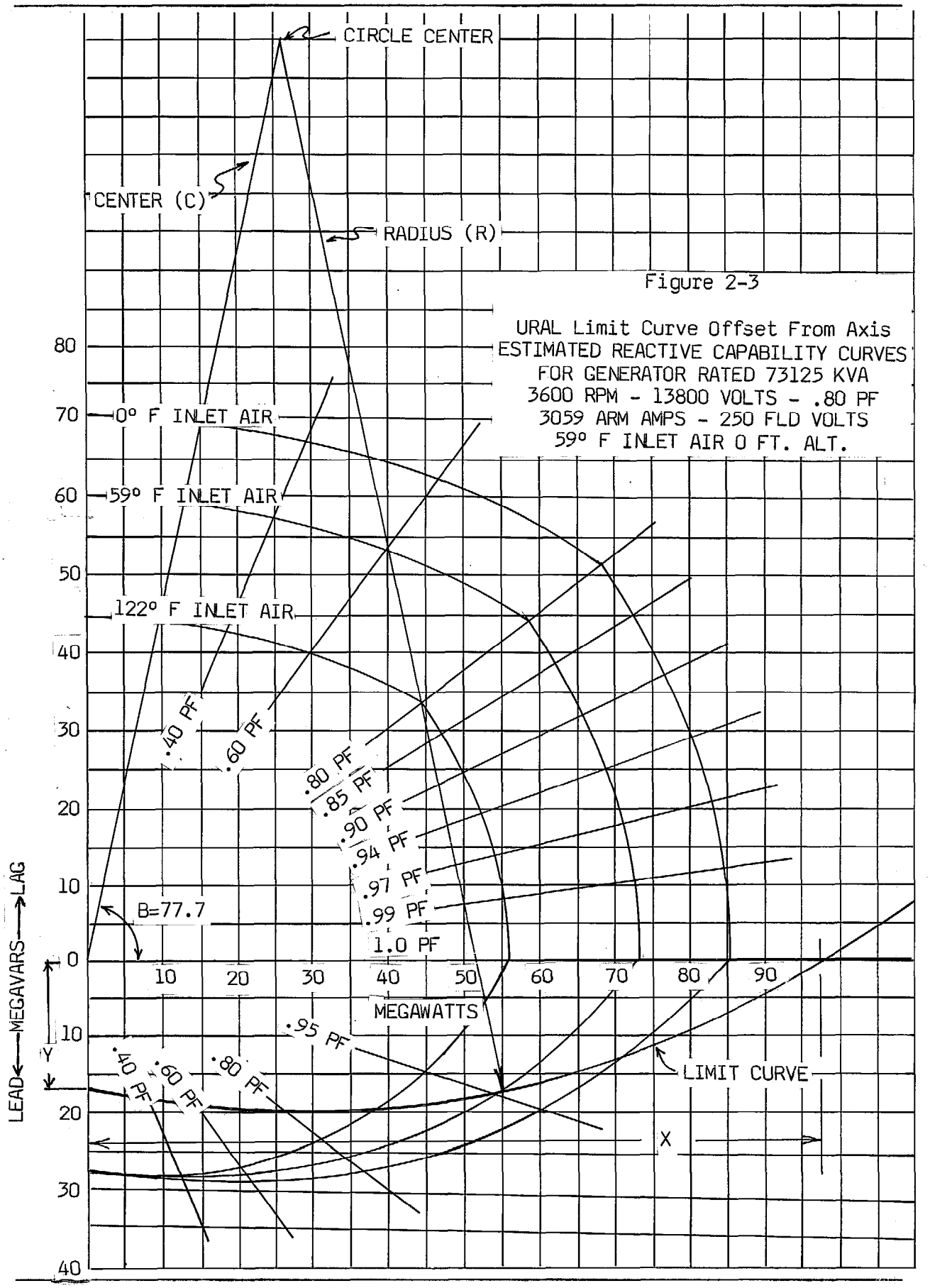

CIRCLE CENTER
CENTER (C)
RADIUS (R)
URAL Limit Curve Offset From Axis
ESTIMATED REACTIVE CAPABILITY CURVES
FOR GENERATOR RATED 73125 KVA
3600 RPM - 13800 VOLTS - .80 PF
3059 ARM AMPS - 250 FLD VOLTS
59° F INLET AIR 0 FT. ALT.
0° F INLET AIR
59° F INLET AIR
122° F INLET AIR
.40 PF
.60 PF
.80 PF
.85 PF
.90 PF
.94 PF
.97 PF
.99 PF
1.0 PF
.95 PF
B=77.7
MEGAWATTS
LIMIT CURVE
LEAD ← MEGAVARS → LAG
X
Y
80
70
60
50
40
30
20
10
0
10
20
30
40
50
60
70
80
90

Figure 2-3

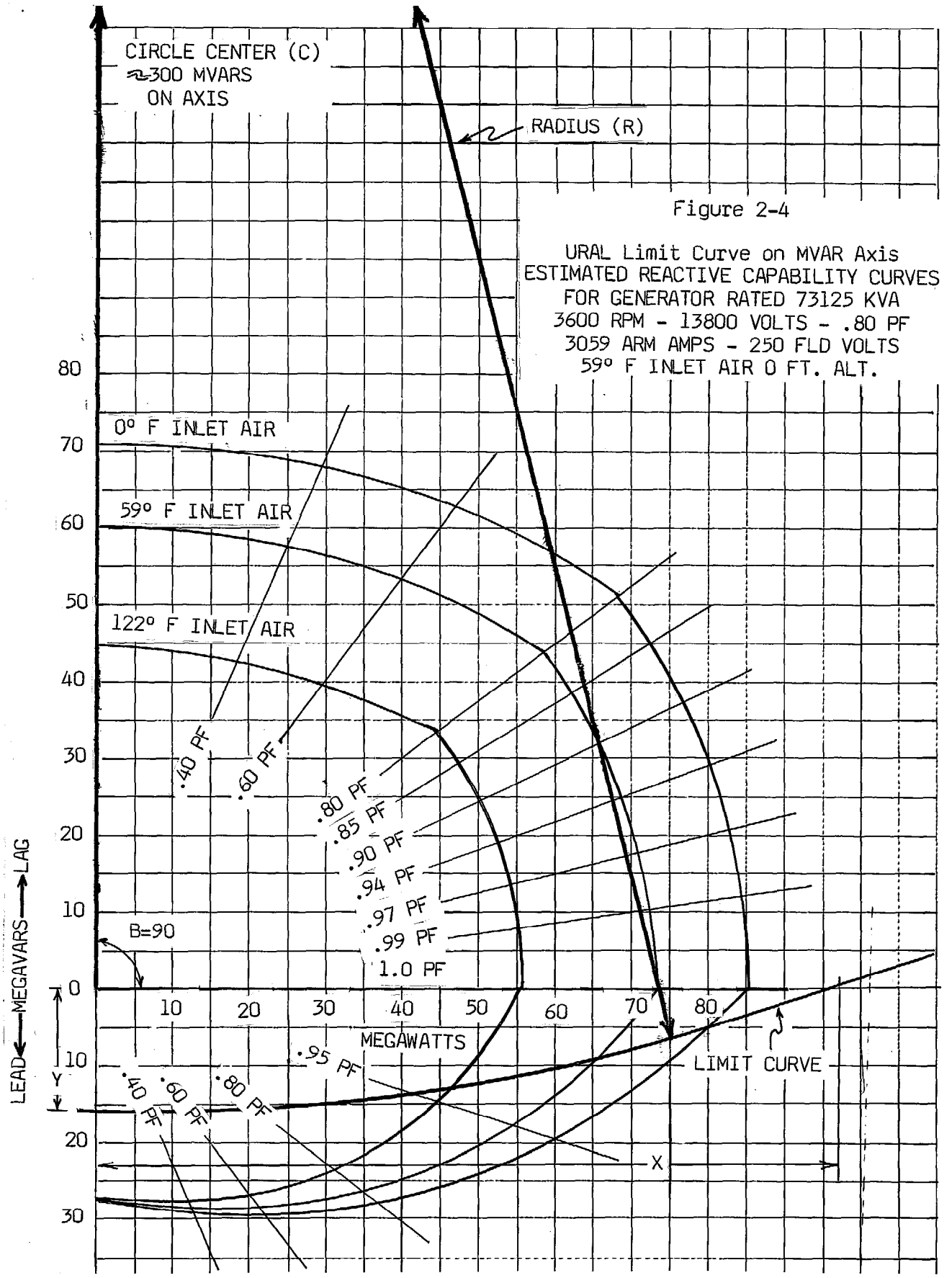

CIRCLE CENTER (C)
≈300 MVARS
ON AXIS
RADIUS (R)
URAL Limit Curve on MVAR Axis
ESTIMATED REACTIVE CAPABILITY CURVES
FOR GENERATOR RATED 73125 KVA
3600 RPM - 13800 VOLTS - .80 PF
3059 ARM AMPS - 250 FLD VOLTS
59° F INLET AIR 0 FT. ALT.
0° F INLET AIR
59° F INLET AIR
122° F INLET AIR
.40 PF
.60 PF
.80 PF
.85 PF
.90 PF
.94 PF
.97 PF
.99 PF
1.0 PF
B=90
LEAD ← MEGAVARS → LAG
MEGAWATTS
.95 PF
LIMIT CURVE
Y
X
80
70
60
50
40
30
20
10
0
10
20
30
40
50
60
70
80

Figure 2-4

NOTE: For 1 Phase or 3 Phase Sensing

When the center is on the ordinate, the angle B is 90°; the above equations simplify to:

② $$C = \frac{X^2 - Y^2}{2Y}$$

$$R = \frac{X^2 + Y^2}{2Y}$$

In the D235T8 example shown in Figure 2-4, it is desired to have the limit characteristic curve pass through the under-excited axis at 16.5 MVARS (.6 X 27.5) or .186 PU (16.5/88.5) and through the unity PF axis at 97.3 MW (1.1 X 88.5) or 1.1 PU.

$Y = .186$ PU $\qquad X = 1.1$ PU

For 3 Phase Sensing

The following method is used to offset the center. Using a compass, construct an arc from the X value using a value of approximately 2.0 PU. (So that the arc does not leave the graph). Place the compass on the Y value and draw an arc that intersects the lst arc. This is the center. Measure the angle B.

B = _____

For the Example Fig 2-3 B = 77.7

$$C = \frac{1.1^2 - .186^2}{2(1.1\ \cos(77.7) + .186\sin(77.7))} = 1.412$$

$$R = \frac{\sqrt{(1.1^2)^2 + 2(1.1 \times .186)(1.1^2 + .186^2)\sin(155.4)}}{2(1.1\cos(77.7) + .186\sin(77.7))} = 1.595$$

These values of C and R should be recorded.

B = _____

C = _____

R = _____

If B is less than 90°, which means the center is in the first quadrant, the jumper on 1TB should be from B to C. If the angle is more than 90°, which means the center is in the second quadrant, the jumper should be from A to B.

1TB = ___B___ to _____

For Single Phase Sensing (or for center on vertical axis)

No offset of the center can be made, therefore, the equations 2 must be used to set C and R. However, the URAL board can only be set up to approximately 4.0 PU. Therefore, if C and R exceed 4.0, pick a smaller value of X or Y.

For the Example Fig. 2-4

$$C = \frac{1.1^2 - .186^2}{2 \times .186} = 3.16$$

$$R = \frac{1.1^2 + .186^2}{2 \times .186} = 3.35$$

These values of C and R should be recorded:

1TB = B to C

C = _____

B = 90

R = _____

Voltage Matching Checks (Sh. 05D, 05E, 05C of Elementary)

Connect a 115 VAC source to the VM signal conditioner transformer. (Sh. 05D). Disconnect PT feed.

Remove the AC Sensor PWB (for generator volts) sheet 05A.

Connect the NEG output of the DC SVA Counter PWB pins 27 and 28 to the VM SVA Control PWB pins 25 and 26. This simulates the-VT signal to the voltage matching circuits.

Energize the 115 VAC source and close 52Z-1X. At this point, relay 60GL should energize, indicating that the generator voltage is lower than the line volts.

Energize 83SR1

The AC SVA should begin to increase. Raise the DC SVA until -VT = -VL. At this point, the AC SVA should stop increasing. Raise the DC SVA further and the AC SVA should decrease.

Current Transducer (Sh. 04B)

1. Measure the input to the transducer at FIT-6 to FIT-8. It should be 0.0 mv.

2. Measure the voltage at FIT-4 to FIT-5. It should be 0.00VDC. Adjust to zero with the BAL potentiometer.

3. Remove the shunt leads at FIT-6 and FIT-8 and connect a millivolt source (0 to 100 millivolts) to FIT-6 and FIT-8. Measure the input mv.

4. Set the input to the millivolts equal to AFFL (on Sh. A04 note J) and measure the output at FIT-4 to FIT-5. Adjust the output to get the required volts (Sh. A04 Note J) with the GAIN potentiometer. The tolerance should be - 0.01 VDC.

   mV_{in} = _____ $Volts_{out}$ = _____

5. Set the input to 0.00 mv.

6. Connect a milliammeter (0-20MA) between FIT-3 and FIT-5.

7. With zero input, the transducer MA output should be 4.00 ma $\pm$ 0.02 ma. Adjust with potentiometer 5P (Zero) on the PS/MA board in the transducer.

8. Set input to 100 mv and the output should be 20.0 ma $\pm$ 0.3. Adjust with potentiometer 4P (SPAN).

9. Repeat steps 5 and 6 until both the SPAN and ZERO are set correctly.

CHAPTER 3
OFF-LINE CHECKS

OFF-LINE ADJUSTMENTS

The initial tests performed previously will allow the startup of the TG set with confidence and eliminate the need for many adjustments. The off line adjustments that follow will allow the initial startup, checkout and adjustment of the regulators and firing circuits. Any circuits not previously checked for proper operation will be checked in this section.

****
* CAUTION *
****

DO NOT OPERATE GENERATOR UNLESS ALL METERING AND PROTECTIVE RELAYING HAS BEEN CHECKED INITIALLY AND IS IN SERVICE.

Off-Line Checks: (Require approximately 8 hours for completion)

_____1. Check that 41AC Exciter Feeder Breaker is closed. (Or that the field breaker 41 is closed).

_____2. Check that 83SR "TEST" Switch is in MANUAL, and that 43SR/CS is in AUTO.

_____3. Check that 120VAC and 125VDC control power is on, and that all generator alarms are cleared.

_____4. Rack up 52G Generator Breaker. (Do not rack while unit is running).

****
* CAUTION *
****

DO NOT ENERGIZE 52G USING THE TEST PLUG WHILE IN THE DC REGULATOR. EXCITATION WILL GO TO 75% OF RATED VALUE, AND DAMAGE TO THE GENERATOR WILL RESULT.

_____5. Check that 64F Field Ground Relay is energized.

_____6. Check that all control fuses are in place.

Regulator Compartment:

a - FU4 and FU5 (for BFE Bridges) or 1FU and 2FU (for 3S7932RA132 bridges) - +36 volts and -18 volts supply to each bridge

b - F1FU and F2FU - Field Volts Supply

c - C3FU and C4FU - Field Flashing Supply (MS7001 only)

d - C1FU and C2FU - 125 VDC Control

e - C6FU - 64F Control - 120 VAC supply for Field Ground Detector

f - FU-1A and FU-2A - PT2 Voltage Supply

g - PT drawers are fully closed

Generator Panel;

a - FU-10 - Line PT Supply

b - FU-4 - 125 VDC Control

c - FU-5 - 86G Control

Generator Breaker Compartment;

a - FU-5 - Close Control

b - FU-6 - Open Control

_____7. Connect an oscilloscope across the Field Voltage Transducer Output (or field).

_____8. When unit reaches _____% speed, 41F Field Flashing Contactor should pick up. Pre-positioning of the AC SVA (and DC SVA) should take place at this time. Quickly check this:

a) AC SVA. 94Z-1 should pick-up. The output of AC SVA Counter Board at O-(+)10 test point should count up or down until it equals the voltage at T-REF (2TP) of the AC SVA Control Board (C03).

b) DC SVA. 94Z-2 should pick-up. The output of DC SVA Counter O-(+) test point should count up or down until it equals the voltage at T-REF (2TP) of DC SVA Control Board (D03).

"Flashing" voltage and current should be at least_____ of VFNL and AFNL respectively. (75-100% for MS5001, MS6001 and 5-10% for MS7001)

_____9. Most important! Check field waveforms per oscilloscope. You should see no "Jitter" from SCR firing (period between pulses of field voltage waveshape = 5.56 msec = 60 cycle). See Figure 3-1 and troubleshooting. Very important! Refer to pages 3-7 through 3-13 for troubleshooting. If the field voltage waveshape is satisfactory at this point, continue with the sequence.

_____10. When rated speed is reached and no problems have occurred, adjust 90R5 to attain rated generator KV. Record the following; KV = _____, Fld volts = _____, Fld Amps = _____, PPT Secondary Volts = _____, (SC8 - SC10). Synchronizing input (RECP 18 - RECP 29) $-V_F$ (2TP Inner Loop) = _____. PT Volts = _____.

PT2 L-L volts; 1PT - 2PT = _____

1PT - 3PT = _____

2PT - 3PT = _____

SC3 - SC4 = _____

SC3 - SC5 = _____

SC4 - SC5 = _____

Signal Conditioner 3 phase output;

SC18 - SC20 = _____

SC18 - SC22 = _____

SC20 - SC22 = _____

PT2 Phase Rotation;

1PT - 2PT - 3PT = _____ - _____ - _____

SC3 - SC4 - SC5 = _____ - _____ - _____

PPT Phase Rotation;

41AC(1) - 41AC(3) - 41AC(5) = _____ - _____ - _____

PPT Secondary L - L Volts;

41AC(1) - 41AC(3) = _____

41AC(1) - 41AC(5) = _____

41AC(3) - 41AC(5) = _____

_____11. Check that the currents are balanced in each line phase of the power Rectifier Bridge. Check at rectifier bridges for multiple bridges

41AC(Phase A) = _____

41AC(Phase B) = _____

41AC(Phase C) = _____

Check the LED's on each bridge (with disconnects) to ensure uniform brightness.

If currents are not balanced, check for the absence of a gate signal (use a scope), a defective cell (use a scope), a blown fuse (use a meter across fuse), or phase reversal (use scope to check phase sequence).

_____12. Check all the signals of the Synchronizer and Reference Phasor Boards.

(Pages 3-14 and 3-15).

NOTE: If it is necessary to adjust, do not adjust while the generator is running unless proper precautions are taken.

_____13. Check the output of the Trigger Generator Boards (Page 3-15).

_____14. Check the output of the Power Supply Module per Table 3-1, Pages 3-15 and 3-16.

_____15. Calculate and set proper operation of the DCPT Field Voltage Transducer (page 3-16).

____16. Calibrate the proper settings of the Inner Loop Regulator Board (Page 3-17).

_____17. Calibrate the AC/DC Board.(Page 3-18)

_____18. Check the settings of the Field Overvoltage Board (Page 3-19).

_____19. Check Signal Conditioner (Page 3-19).

_____20. Check for proper input settings of Volts/Hz Trip Boards (Page 3-20).

_____21. Check for proper output of the AC Voltage Sensor Board (Page 3-21).

_____22. Check operation of the AC SVA (Page 3-21).

_____23. Connect DVM to 5TP on AC/DC Board to act as a Transfer Voltmeter.

_____24. Connect the oscilloscope to the $-V_F$ and $-V_T$ signals:

$-V_F$ ----- 2TP of Inner Loop Regulator

$-V_T$ ----- 2TP of AC Volt Sensor

_____25. Check that the Impedance Compensator Board (1M3E) and URAL Board (1M3A) are withdrawn from their slots so that pins are not connecting. Also remove the V/Hz Regulator PWB if supplied. Disable the Voltage Matching circuits (52Z-X).

_____26. Thru adjustment of the AC SVA, try to zero the DVM connected to 5TP of the AC/DC PWB, or get as near to zero as possible.

_____27. With DVM at zero (or as it passes by zero), turn 83SRX "TEST" switch to Auto while observing the field voltage, the $-V_T$ signal, and the $-V_F$ signal for stability. If unstable, transfer each to Manual. Consult troubleshooting (Pages 3-11 through 3-13).

_____28. Check optimum regulator stability (Page 3-21).

_____29. Check the range of the AC SVA (Pages 3-21 and 3-22).

_____30. Check the operation of the Impedance Compensator Board (Page 3-22).

_____31. Check the operation of the URAL Board (Page 3-22 and 3-23).

_____32. Check the operation of the Volts/Hertz Regulator Board (Page 3-23).

_____33. Check the operation of the Voltage Matching circuits (Page 3-23).

_____34. Check the calibration of the Field Current Transducer (Page 3-24)

(Must be in DC Reg).

____35. Check the operation of the Field Temperature Indicator.

Measure the actual field volts and divide by the field current. The temperature meter should read:

$$R_{calc} = \frac{\text{Volts}}{\text{Amps}}$$

$$T = \frac{R_{calc}}{R_{25c}} \times 259.5 - 234.5$$

_____36. Check the operation of the Auto Tracking function (Page 3-25).

_____37. Set the On-Line Pick-up Adjuster 90R1 (Page 3-26).

_____38. Set 90P (Page 3-26).

_____39. Set 90R5 to hold rated terminal volts in DC Reg. (Sh. A04 note E2C).

_____40. Check that 83SRX TEST Switch is in Auto.

_____41. Verify operation of 43SR/CS and the Transfer Voltmeter at the Generator Panel, through transfers and raising and lowering of the AC voltage Controllers.

_____42. This completes the Off-Line checks. Read On-Line checks before continuing.

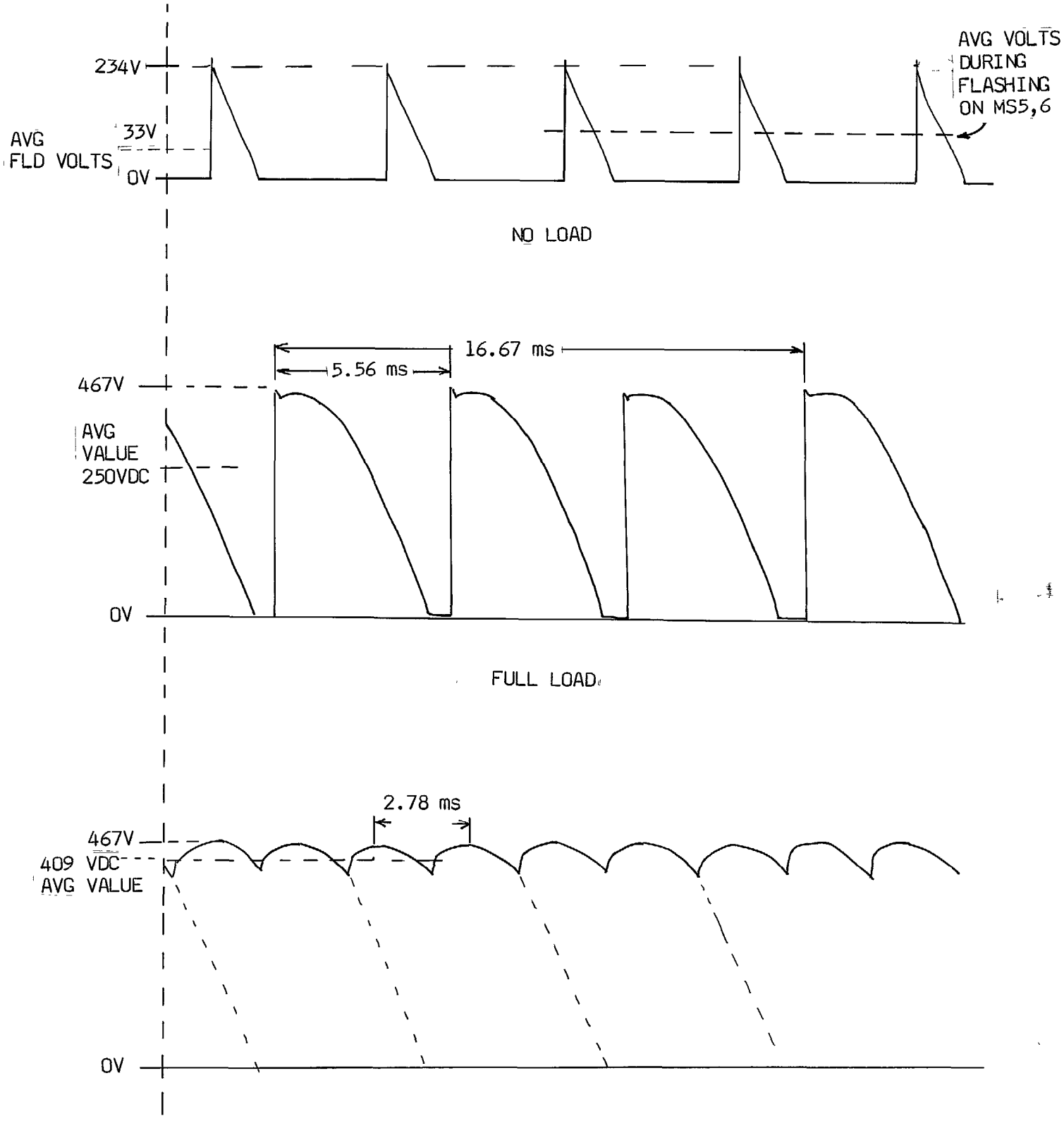


Figure 3-1
Field Voltage Waveforms with 330 VAC PPT

Rectifier Bridge Trouble Shooting Procedure

1. Make sure that the operating condition during trouble shooting is safe from a personnel and equipment standpoint. The rectifier bridges are capable of operating continuously at peak load at maximum ambient temperature for that application. If a component fails, the other diodes/SCRs may pick up that components load. You must operate at the lowest voltage that sustains the indications.

2. Transient phenomenon must be recorded with a chart recorder or memory scope or converted to a continuous event. Do not operate a ceiling (maximum excitation) for more than 15 seconds (with a 5 minute cool down).

3. Start checking by discarding all assumptions. Do not assume PPT is phased correctly by the factory. Do not assume there is no ground on the secondary of the PPT. Do not assume diodes/SCRs are OK, etc.

4. Once the basic power circuits are fully checked, recheck the firing circuits (Synchronizer, Reference Phasor, Trigger Generator, Inner Loop, DCPT). Pick one point to start and trace signal flow. Look for the quality of the signals, but do not discard small pulses or pips which may cause false triggering.

5. Recheck 0 VDC bus and all grounds.

6. Analyze all events that caused the problem. Was a PWB inserted then removed? Does it only happen when a PWB is on the extender, or when an instrument is connected to a test point?

7. More than one problem may exist, so try to eliminate all but essential functions, and try to isolate the problem by trying different solutions.

8. Look for very high frequency noise (MHZ) on firing and regulator circuits as well as on power supplies. Indication would be a fuzzy line when it should be DC.

9. Record methods used and results to facilitate factory (if needed).

10. When trouble shooting DS3820BFE Rectifier Bridges, do not disconnect only one plug connector to the gate driver cards; disconnect all three. This will prevent cross coupling of firing pulses.

Bridge Output Troubleshooting - (Waveforms shown below may not be exact)

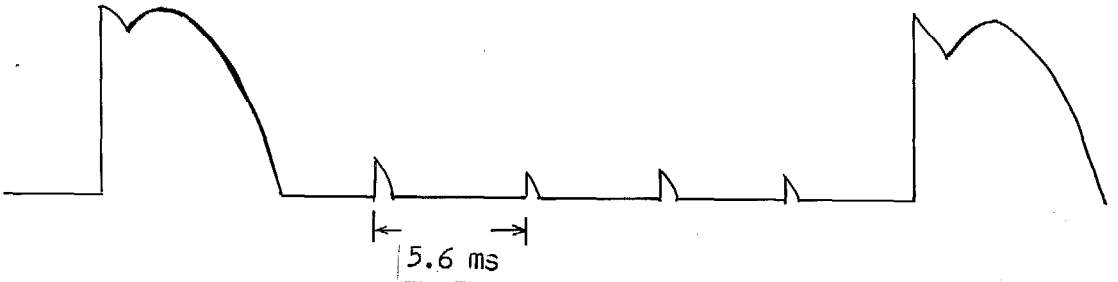
1.



Waveform uneven

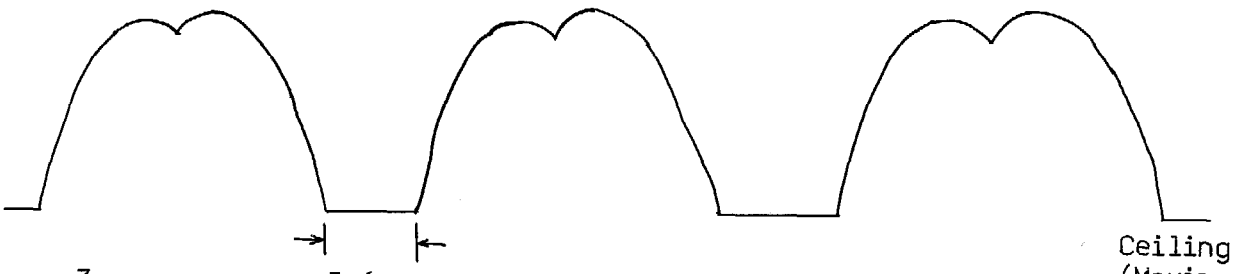
Cause - PPT secondary volts unbalanced or Reference Phasor not set up correctly (Page 3-14) or defective Trigger Generators. (Page 3-15). If oscillation of pulses occur that is not synchronous with line, then DC regulator may be oscillating.

2.



Cause - Extra firing pulses into gate drivers (Page 3-15). Look for small pulses on the output of the Trigger Generators at the same time as other Trigger Generators start their pulses. Solution may be to rewire to reduce pickup.

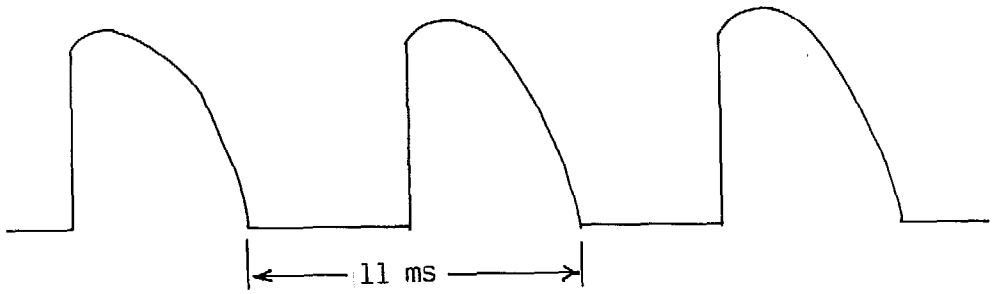
3.



Ceiling (Maximum) Excitation

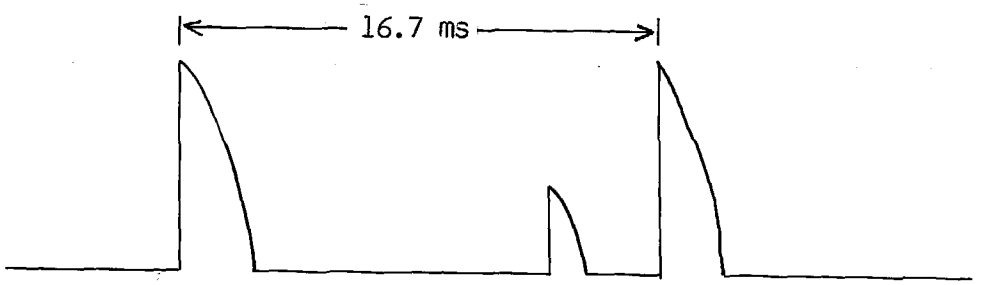
Cause - Self firing of SCRS - Free wheeling diode may be damaged (open). Damage may occur if unit is run in this condition for more than 30 seconds. Firing pulses may or may not be present at Gate Drivers. Check entire firing circuit (Pages 3-14 and 3-15).

4.

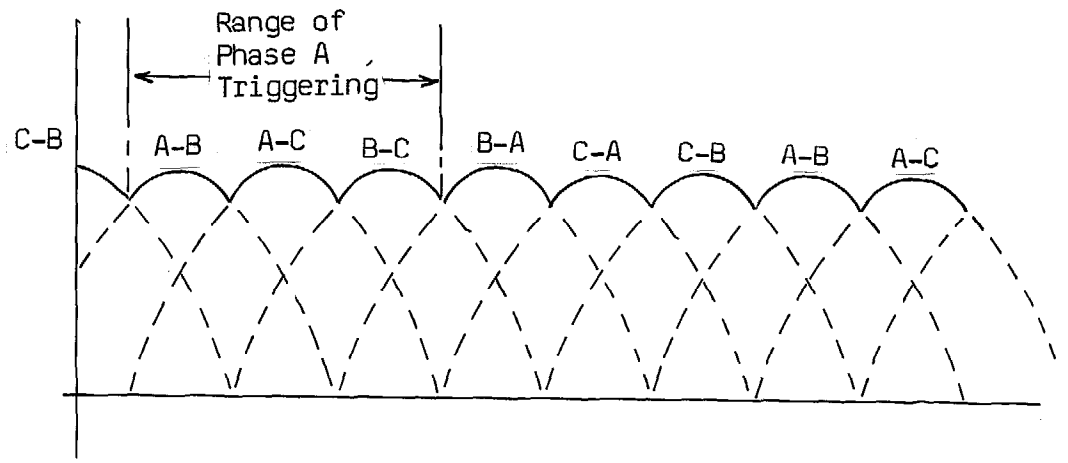


Cause: Missing gate driver pulses

5.



Cause: Blown fuse



Troubleshooting

Many of the regulator functions are on separate printed wiring boards. You can replace or troubleshoot some of these boards while the generator is still on line. Board replacement can be made immediately upon discovery of the cause of failure and then the determination of whether to return the board for repair or to replace with a new board can be made. Field repair of printed circuit boards is difficult and should only be attempted by qualified personnel, remembering that damages resulting from this will not be covered by warranty.

The following boards may be withdrawn from the PWB rack assembly with the DC regulator in control.

1. AC Voltage Sensor
2. AC Regulator
3. Impedance Compensator
4. AC SVA Counter
5. AC SVA Control
6. Volts/Hertz (Optional if furnished)
7. URAL (Optional if furnished)
8. Voltage Matching (Optional if furnished)

****
* *
* CAUTION *
* *
****

THE FOLLOWING BOARDS MUST NOT BE WITHDRAWN WHILE THE EXCITATION SYSTEM IS SUPPLYING FIELD CURRENT TO THE GENERATOR REGARDLESS OF WHICH REGULATOR IS IN CONTROL.

1. AC/DC
2. Synchronizer
3. Reference Phasor
4. Trigger Generators
5. Inner Loop Regulator
6. Field Overvoltage

The following boards may be withdrawn from the PWB rack assembly with the AC regulator in control.

1. DC SVA Counter and Control
2. Transfer and Tracking

TROUBLESHOOTING GUIDE

Trouble	Possible Cause
1. No buildup when flashed.	A. Flashing contactor not picking up.
	B. Flashing voltage not available.
	C. Flashing circuitry open.
	D. Field circuitry open/shorted
	E. Blocking diode open.
	F. Flashing fuses blown.
2. Field Voltage goes too high or low during flashing.	A. Start-up resistor wrong value.
	B. 90R5 incorrectly set.
	C. DC regulator problems.
	D. Shorted flashing diode.
	E. Open/shorted field sensing input (DCPT).
	F. Firing circuits (-ER Signal).
	G. Noise in firing circuits.
3. Partial buildup but decays after flashing terminates.	A. Flashing current too low.
	B. DC regulator failure.
	C. Power supply failure.
	D. All SCR'S not firing.
	E. Anode supply voltage not available to SCR Bridge. 41AC open.
	F. DC regulator set too low.
	G. Blown fuse in rectifier bridge.
	H. Inner Loop Reg. or DCPT failure(high)

Trouble	Possible Cause
4. Terminal voltage goes to ceiling after flashing (DC regulator)	A. DC regulator set too high.
	B. 52GX-1A energized.
	C. Inner Loop Regulator failure.
	D. AC/DC gate failure.
	E. Transducer failure (low).
	F. Noise in firing circuits.
5. Terminal voltage goes to ceiling in AC regulator	A. Loss of PT signal, -VT signal.
	B. Impedance compensator defective.
	C. AC regulator problems.
	D. Min. setpoint too high.
	E. 90P not set correctly.
	F. AC/DC PWB not selecting AC Reg.
	G. Overexcitation PWB failure.
	H. Arcing or corona on HV bus.
6. Field voltage wave shape not normal. (See Figure 3-1).	A. SCR, diode failure.
	B. Fuse blown.
	C. Phasing error.
	D. Ripple on AC/DC sensing (-ER).
	E. Synchronizer not adjusted.
	F. Ref. phasor not adjusted.
	G. Trigger gen. or gate driver failure
	H. System unstable.
	I. DC regulator oscillations.

7. Terminal voltage goes to zero in AC regulator only.	A. Sensing voltage too high.
	B. AC Regulator range wrong.
	C. AC Regulator failure.
	D. AC/DC Gate PWB failure.
8. Excitation System unstable.	A. Regulator gain set too high (AC or DC)
	B. Regulator not stabilized (AC only) Adjust lead/lag so K5=1.0, K4=.1
	C. Ripple on sensing.
	D. Power supply trouble.
	E. Regulator failure.
	F. MW variations in turbine.
9. Slow response (AC regulator)	A. Regulator gain set too low.
	B. Regulator not stabilized properly. (lead/lag adjustment).
10. Poor regulation (AC regulator)	A. Regulator gain too low.
	B. Impedance compensator not adjusted properly.
	C. Sensing distorted with load.
11. SCR stays on or output latches up.	A. Caused by SCR latching on after being turned off. Current in SCR continues to flow.
	B. Failure of Free Wheeling rectifier.
	C. Noise in firing circuits.
12. Excitation overvoltage alarm when generator is at rest.	A. Failure of DCPT.
	B. Failure of Inner Loop Reg. PWB.
	C. Failure of Field Overvoltage PWB.

Synchronizer Checks: (Sh. 07A)

Check the signals on the Synchronizer board. Put the oscilloscope on test point 5TP (-E SYNCH). The voltage should be clipped at about ± 10 volts. The signal at 3TP (QA) should lag the signal at 4TP (IA) by 90 degrees.

Reference Phasor Board Checks: (Sh. 07A)

Check the signals on the Reference Phasor board with an oscilloscope. The signal at 2TP (-IC) should lag the signal at 4TP (QC) by 90 degrees. The signal at 3TP (-IB) should lag the signal at 5TP (QB) by 90 degrees.

SYNCHRONIZER, REF. PHASOR & TRIG. GENERATOR BOARD

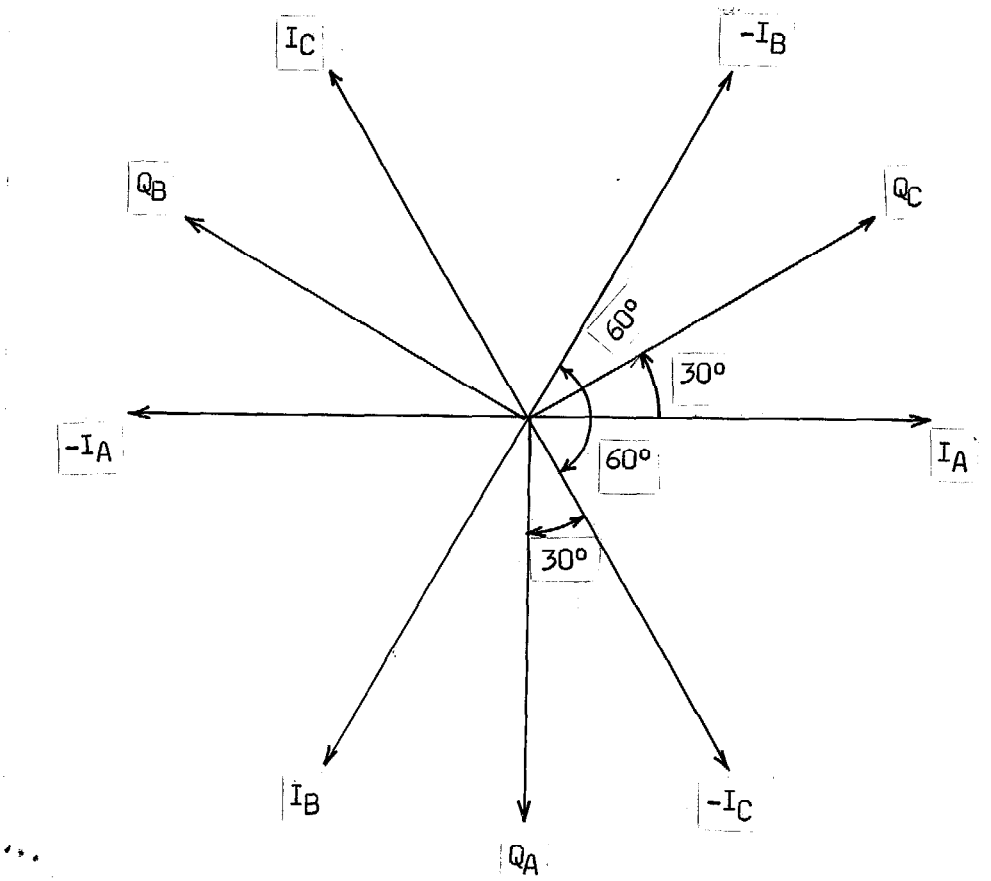


Figure 3-2
Q and I Vectors in Reference Circuit

Signals generated by Synchronizer Bd. and Reference Phasor Bd.

NOTE

If the Q signals shown in Figure 3-2 are not equal in amplitude, you can adjust the amplitude as follows:

1. Q_A ; Adjust potentiometer 3P on the Synchro Board.

2. Q_B ; Adjust potentiometer 6P on the Ref. Phasor Board.

3. Q_C ; Adjust potentiometer 4P on the Ref. Phasor Board.

(There is no adjustment for I_A, $-I_A$, or $-I_C$)

Trigger Generator Board Checks: (Sh. 07B)

Check the output of each Trigger Generator PWB at 5TP (OUT). The output should appear as in Figure 3-3. Each Trigger Generator output should be displaced by 120 ± 2 degrees (5.56 msec at 60 Hz or 6.67 msec at 50 Hz).

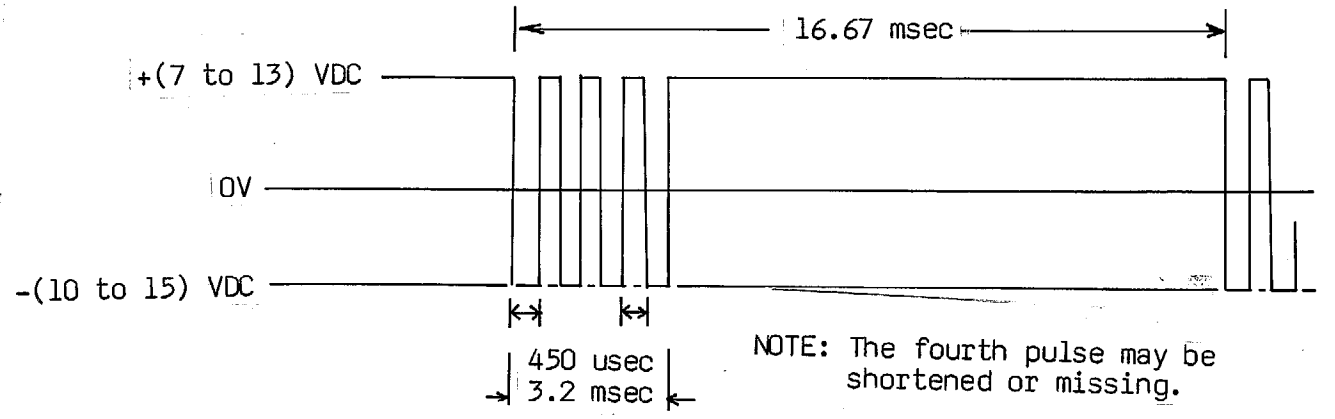


Figure 3-3
Trigger Generator Output (5TP)

Table 3-1
Power Supply Bus Voltages (Sh. 08A)

				PS9 - 10	PS11 - 10
INPUT	+15V Bus	-15V Bus	-24V Bus	36V Bus	-18V Bus
115 VAC	+15.0	-15.0	-24.5	+44.0	-21.0
Tolerance	+.6V	±.6V	±2.4V	±6.0V	±3.0V

If the measured voltages do not correspond to those given in Table 3-1 within the listed tolerances, correct the problem before proceeding. Check taps of primary of transformers.

DCPT Transducer (Field Voltage Transducer) (Sh. 04B)

Final calibration of the transducer must be performed off line. Refer to sheet 04B of the elementary to see which functions are connected to the transducer. The following options may be supplied with the transducer: Field Temperature Indicator, Field Voltmeter (4-20ma) and Remote Field Voltmeter. If these options are not supplied, skip the applicable steps.

1. Measure the field voltage with a DVM. Make sure regulator is in DC mode.

2. Calculate the output voltage of the transducer by the following equation:

$$V_{out} = V_{field\ input} \times \frac{V_{out\ Cal}}{V_{field\ Cal}}$$

where $V_{out\ cal}$, $V_{field\ cal}$ are given on sheet A04, Note H.

V_{out} = _____ X _____ =

3. Measure the output voltage at FVT-4 to FVT-5.

$V_{out\ (ACT)}$ = _____

4. The tolerance should be as follows:

Standard (No options)	= ± .05V
Field Temp	= ± .01V
Meters	= ± .01V

If the output does not correspond to the calculated value within the tolerance, adjust by the following method:

A. Adjust the output of the transducer to the correct value with $\overline{90R5}$.

B. Measure the field voltage with a DVM and adjust to the correct (previous) value with the GAIN potentiometer.

C. Recheck the transducer output.

5. (For meter outputs) measure the MA output at FVT-3 to the meter lead (USE MA SCALE 20 ma).

MA_{out} = _____

6. Calculate the MA output by the following equation:

$$MA_{out} = 4.0 + \frac{(V_{in}}{V_{scale}} \times 16)$$

where V_{in} = field voltage input

V_{scale} = full scale voltage of meter

$MA_{out} = 4.0 + (_____ \times 16) = _____$

7. The output MA should be within ± 1% at full scale. At no load (FSNL) the output should be ±.02 MA.

8. Adjust 4P (SPAN) to get the correct value if necessary. If 4P is adjusted, the zero must be recalibrated by the following procedure.

   A. Open 41AC to remove excitation.

   B. Check MA out. It should be 4.00 ma ± .03MA. Adjust with 5P (Zero) to this value.

   C. Reclose 41AC. Wait until the field voltage has stabilized and measure field volts. It should be the same as before.

   D. Recheck MA out. If the field voltage has changed, recalculate ma output (as above). Adjust 4P (SPAN) to get within tolerance (±.02MA).

   E. Repeat steps A-D until tolerance is met.

Inner Loop Regulator Checks:(Sh. 06A)

With the generator operating at rated speed and rated terminal voltage, measure the field voltage with an analog VOM. If the signal oscillates, reduce the Inner Loop Gain by adjusting 6P on the Inner Loop Regulator board faceplate.

Measure the voltage at 2TP (-VF) on the Inner Loop Regulator PWB. Adjust the potentiometer 1P (GAIN) to get the proper -VF signal for the actual field voltage. Use the values on Sh. A04 - E1A for the calibration.

$$Vf = \frac{V_F \text{ (Actual)}}{V_F \text{ (Calibration)}} \times 5.0 \qquad \text{(Within } \pm .01 \text{ VDC)}$$

To set the correct value, adjust -VF to the calculated value with 90R5. Measure the field voltage with a DVM adjust the GAIN potentiometer 1P until the field voltage is the correct value.

AC/DC PWB Calibration (Sh. 06A)

Set the 90R5 to minimum. Pins 17 and 18 on the AC/DC PWB should be 0 VDC. Use the MIN potentiometer 2P to adjust the field voltage to the MIN value given on Sh. A04 - E2A.

Do not adjust the SPAN potentiometer at this time.

NOTE: At this time determine if the calculated NO-LOAD field volts, VFNL, agrees with the measured value across the field. If it does not, the field overexcitation values must be changed (use linear ratio) and the settings at 90R5 and 90R6, as well as the MIN and SPAN may need to be changed (if the difference is more than 10%). Investigate if change is too great.

Estimated Rotor Temperature = _____

Field Current to hold Rated Gen. Terminal volts (AFNL) = _____

Field Voltage to hold Rated Gen. Terminal Volts (VFNL) = _____

Field Resistance at above temperature = $\frac{V_F}{I_F}$ = _____

R_f @ 25°C = R_f @ T X $\frac{295.9}{234/5+T}$ = _____ check specified value on Sh. A04

R_f @ 125°C = R_f @ T X $\frac{359.5}{234/5+T}$ = _____ check specified value on Sh. A04

The following equations should be used in calculating the new settings.

Over Excitation

OFF-LINE = 1.15 X AFNL X R_f @ 125°C + 10V

ON-LINE = AFPL (Peak Load) X R_f @ Max Temp + 10V

DC Regulator

MIN = .60 AFNL X R_f @ 25°C

SPAN = 1.1 X AFPL (Peak Load) X R_f @ Max Temp.

90R5 = AFNL X R_f @ 25°C

90R6 = .75 X AFFL X R_f @ 125°C

Field Overvoltage Checks (Sh. 10A)

The field overvoltage protection must be calibrated using the actual generator values of field current and resistance. The generator characteristics may cause a 15% deviation from the calculated values. If the actual values are more than 5% above the calculated values, recalibrate the settings using the actual field values.

If a turbine trip is not wanted, open the 59E-3X (Sh. 11A) trip circuit. Lower 90R5 to minimum and wait until the generator voltage reaches the minimum. Connect the oscilloscope to 2TP on the Field Overvoltage PWB. Connect the DVM to 2TP (-VF) on the Inner Loop Regulator PWB. Quickly raise 90R5 until 2TP on the Overvoltage PWB just goes to zero. Record field volts and -VF signal.

The 41AC breaker should trip within 10 seconds. The LEDs will flash. If the value of -VF and field voltage do not agree with the calculated values, the setpoint must be readjusted.

Signal Conditioner Checks (Sh. 05A)

For 1 Phase Sensing: At rated generator terminal volts, measure the PT voltage at SC-3 to SC-5.

PT Volts_____

Measure the secondary volts at SC-15 to SC16 and SC16 to SC17.

SEC 1 = _____

SEC 2 = _____

The secondary voltage should be between 16.5 and 19.3 volts AC. If it is not, adjust the taps of the transformer 1T (use SC4 or SC6). The secondary should be nominally 18V RMS at rated value.

For 3 Phase Sensing: At rated generator terminal volts, measure the PT voltages at SC3, SC4 and SC5 (Line to Line).

Phase 1 SC-3 = _____ Phase 2 SC-4 = _____ Phase 3 SC-5 = _____

Measure the secondary voltages to 0V com (SC24) at SC-15, SC-16, SC-17.

E Phase 1 SC-15 = _____ E Phase 2 SC-16 = _____ E Phase 3 SC-17=_____

If the voltages are not 17 VRMS (± 1V), remove the 3 leads on 4TB-6 and connect 1TB, 2TB and 3TB points 5 (for a higher voltage) or points 7 (for a lower voltage). See GEK-73812 for details of the module.

V/Hz Trip Setting (Sh. 10C):

Place Volts/Hz Input Board (1M4G) on the extender board. With machine at rated voltage, perform the following: Measure the output of amplifier 3A (2TP) on the V/Hz Trip Input PWB. This output is proportional to the V/Hz of the input signal. This should be -5V. Adjust, if necessary, using potentiometer 2P on this board.

2TP(3A) = _____

Ascertain that the voltage at test point 3TP is 11V ± 1. If not, set to 11V, using potentiometer 3P.

3TP = _____

Operate the Test Switch on the V/Hz Trip Output board. The TEST/TRIP LOCKOUT light on the Output board should flash. This should not change the output of amplifier 3A on the V/Hz Trip Input board. If it does, adjust potentiometer 1P on the Input board to whichever extreme does away with the change.

Using 90R5 (or DC SVA) slowly raise generator volts and observe the voltage at 4TP on the Input board. This is the output of amplifier 4A, which should be at negative saturation (about -14V) until the output of amplifier 3A reaches -5.5V. This is equivalent to 1.1 PU V/Hz. If amplifier 4A starts increasing at a lower value of V/Hz, increase the voltage at 3TP, using potentiometer 3TP. If the reverse is true, decrease the voltage at 3TP.

If possible, raise DC SVA (Generator KV)* - until output of amplifier 3A (2TP) reaches -5.6 VDC. Relay 59VH-1X should pick up. Period of oscillations at 6TP on the V/Hz Input Board should be 18.0 msec. 180 seconds later the trip LED should flash indicating trip. (If not, at first outage, perform V/Hz trip setting adjustments as per write-up in the Pre-Excitation Checks).

NOTE: Do not "Overvoltage" the generator during these checks.

Remove V/Hz Trip Input Board from the extender and place back in its slot.

Turn off Test/Trip Lock-Out Switch and Reset Flashing LED.

A further check may be made by increasing the generator terminal volts 12% above rated. At this point 59VH-1X should energize and the annunciator should pick up. After 3 minutes, the Turbine/Generator should trip. (Remove Overexcitation relay 59E-2X to prevent overvoltage trips.

When a restart is made, after the TG is at rated speed, increase the generator* volts by 20%; 2 seconds later the TG should again trip.

* Or lower the Turbine Speed

AC Voltage Sensor Board Checks (Sh. 05A):

Measure the generator PT secondary voltage at rated generator line voltage. (Adjust with 90R5). Measure the voltage at 2TP (OUT) on the AC Voltage Sensor PWB. This is the -VT signal and should be -5.0 VDC at rated terminal volts. Adjust this value to -5.0 VDC by potentiometer 1P (GAIN) (within ± .01 VDC).

AC SVA Operational Checks (Sh.05C):

Exercise the AC SVA with the local pushbuttons and have someone exercise the Raise/Lower switch on the Generator Control Panel. At the MIN limit the SVA output should be 0 VDC. From MIN to MAX the time should be 60 seconds. At the MAX limit, the output should be +10 VDC.

Regulator Stability Checks:

Check the AC regulator stability by introducing an error signal. To do this, 90R1 is adjusted to change the terminal voltage by 5%. Set 90R1 for +2.5 VDC and energize 52GX-1A (must be in AC Reg). The terminal voltage should change 5% in less than 2 seconds with little overshoot and no oscillations (use the -VT signal to observe). See Figure 3-4. Consult the troubleshooting section if this step does not yield satisfactory results. De-energize 52GX-1A. A further test of stability can be obtained by transferring between the AC and DC regulators and back again with an offset in the transfer voltage. Both the AC and DC regulators should be stable in both directions.



Figure 3-4
Waveshape of Terminal Voltage

AC SVA Range Checks (Sh. 05B):

Set the range of the AC regulator by lowering the AC SVA to MIN and then raising it to MAX. Measure the PT signal and the -VT signal. If the terminal voltage range is not satisfactory to the customer, or if the range is not per sheet A04, adjust the range by using the MIN and SPAN potentiometers on the AC Regulator PWB. Set the AC SVA to MIN. Adjust potentiometer 2P (MIN) to the lowest terminal voltage required. Set the AC SVA to MAX. Adjust potentiometer 1P (SPAN) to the highest terminal voltage required. Note 90R1 must be added to obtain the actual ON-LINE range. (Close 52GX-1A to get on line range).

****
* CAUTION *
****

THE GENERATOR TERMINAL VOLTAGE SHOULD NOT CONTINUOUSLY BE EXCEEDED BY 10%, OTHERWISE DAMAGE MAY OCCUR.

MIN VOLTS = _____

MAX VOLTS = _____

Impedance Compensator Settings (Sh. 05A):

Check the operation of the Impedance Compensator PWB. Transfer to DC regulator so that an erroneous output will not cause a change in excitation. Place the Impedance Compensator PWB on the extender. Make sure the jumper is on DROOP (RCC). Measure the voltage at pins 11 and 12 with an oscilloscope. The voltage at pins 11 and 12 should be in phase with the voltage at SC-17 and test point 5TP (Phase 3) on the AC voltage Sensor PWB. Measure the voltage at 9TP. It should be -5.0 VDC with rated terminal volts and no line current. Set with potentiometer 7P. The voltage at 2TP (OUT) should be zero. Use 8P to set this if necessary. Replace the PWB directly into the slot. Transfer to AC regulator and verify that the output of the Impedance Compensator does not affect the AC Regulator. Set the R and X potentiometers to zero. Remove the PWB so the pins are not making contact (after going back to DC Regulator).

URAL Settings (Sh. 05B):

From the pre-excitation checks, record the values of B, R, C and the jumper position.

B = _____

R = _____

C = _____

1TB = _____ to _____

Transfer to DC regulator and insert the URAL PWB on the extender with the generator at rated voltage, adjust the VCAL potentiometer, 2P, until the voltage at 8TP is 5.66VRMS. The voltage at 3TP (OUT) should be at negative saturation (-9 to -14 VDC). If it is not, try turning 1P clockwise.

For 3 Phase Sensing: Check phasing of input voltages with an oscilloscope, per elementary. Using an oscilloscope, place CH1 on 9TP. Adjust the trigger (line synch) until 1/2 cycle is 9 divisions. (1 division = 20 Deg). Place CH2 on 8TP. Adjust 7P until CH2 until CH2 lags CH1 by the angle B.

If B = 90° and the jumper is connected from A to B, and you can not get CH2 lagging CH1 by 90°, try it with the jumper connected from B to C.

For 1 Phase Sensing: Check phasing of input voltages with an oscilloscope, per elementary. Since there is no voltage at 9TP, 8TP should be in phase with pins 15 and 16, and 180° out of phase with pins 13 and 14. The jumper should be on B to C and 6P turned counterclockwise.

Remove the URAL PWB from the extender and place it into rack. The pins should not make contact.

Volts/Hertz Regulator Checks (Sh. 05B):

With the generator at rated speed and voltage, and the DC Regulator in control, insert the V/Hz Regulator PWB into the rack.

Measure the voltage at 2TP (OUT) and adjust to the value given on Sh. A04, note 4, with 1P (GAIN). This sets the reference for the regulator.

Transfer to AC Regulator. The V/Hz Regulator output should not affect the AC Regulator if its output is more negative than the -VT signal (plus the Impedance Comp).

Increase the generator voltage using the AC SVA. When the voltage reaches the PU value given on Sh. A04 (typically 1.09PU) no further increase in terminal volts should be possible.

Voltage Matching Checks (Sh. 05D - 05E):

Ensure that all the Voltage Matching PWB's (VM AC Voltage Sensor, VM SVA Control, (2) PT Transfer and Alarm) are inserted into the rack.

Measure the voltage at 2TP on the PT Transfer PWB. Adjust this voltage to +5VDC using potentiometer 3P (Failure). This indicates input #2 is less than #1 by more than 10% (2% per volt). Measure the voltage at 3TP on the PT Transfer PWB. Adjust this voltage to -1VDC using potentiometer 2P (Control). This indicates that input #1 is less than #2 by more than 2% (2% per volt).

Repeat the above settings for both PT Transfer PWB's.

Measure the input voltage to the VM Signal Conditioner (Line PT output). Adjust the output of the VM AC Sensor PWB using potentiometer 1P (GAIN) until its output is the same percentage deviation from rated as the line PT signal (use 5.0 VDC as rated). As an example, if the line PT output is 116 VAC. 2TP should be set for $\frac{116}{115}$ X 5.0 = 5.04 VDC.

Transfer to AC Regulator and set the generator voltage (PT secondary or -VT) to the line voltage (Line PT secondary or -VL). Use the AC SVA to control generator volts.

Increase the generator volts by 2%. Relay 60GH should energize. Use 2P on PT Transfer #2 PWB to adjust this value.

Decrease the generator volts by 2% below the line volts. Relay 60GL should energize. Use 2P on PT Transfer #1 PWB to adjust this value.

Return the generator volts to the same value as line volts. Both 60GH and 60GL should be de-energized.

Decrease the generator volts to 10% below line volts. Energize 52Z-1X. The AC SVA should begin to increase generator volts. When the generator volts are within 2% of line volts, 60GL should de-energize. When generator volts matches line volts, the SVA Raise/Lower LED's will both flash, indicating a match.

De-energize 52Z-1X and raise the generator volts. Energize 52Z-1X again. The generator voltage should be lowered to line volts.

Field Current Transducer (DCCT) (Sh. 04B)

Final calibration of the field current transducer should be done off line with the DC Reg. in control.

Perform the following:

1. Measure the millivolt (mV) input on FIT-6 to FIT-8.

   mV_{in} = _____

2. Calculate the voltage output from the following:

   $$V_{out} = mV_{in} \times \frac{V_{out\ Cal}}{mV_{in\ Cal}}$$

   Where V_{out} cal and mV_{in} cal are given on sheet A04 Note J.

   V_{out} = _____ X _____ = _____

3. Measure the output voltage at FIT-4 to FIT-5.

   $V_{out\ act}$ = _____

4. The tolerance for the actual depends upon the application
   For Field Temperature Indicators tol = $\pm$ 1%
   For all but the above tol = $\pm$ 5%

5. If the tolerance is not within the specified, adjust GAIN to the correct value.

6. Measure the MA output at FIT-3 to the meter lead (If a meter is supplied).

   MA_{out} = _____

7. Calculate the correct output from the following:

$$MA_{out} = 4.0 + \frac{mV_{in}}{100} \times 16$$

   Where mV_{in} is measured input mV.

8. The tolerance for this value is $\pm$ .02 MA.

9. If the measured is not within the tolerance of the calculated, the output must be adjusted.

   A. If only a meter is used (no Field Temp Ind) Adjust GAIN until the output is correct.

   B. If a Field Temp Indicator is used (or additional meters) use 4P (SPAN) to adjust to the correct value. If this is done, the zero must be rechecked by opening the mV input and checking the MA output (it should be 4.00 ma $\pm$ .03 ma). Repeat MA check until both the ZERO and SPAN are correct.

Auto Tracking Checks (Sh. 06C):

The Auto Tracking function can be exercised while off line by energizing 52GX-1A and selecting the AC Regulator (83SR-1 energized).

When 52GX-1A is energized, 90R1 will cause an increase in generator voltage, the overexcitation setpoints will be changed to the on-line values, the DC SVA instead of 90R5 will control the DC Regulator and the VAR-PF controller will be enabled.

The DC SVA output will track the output of the AC Regulator to keep the TVM nulled. The actual raise and lower signals will occur after a delay of 30 seconds. When the Raise/Lower signal stops, the TRUNB signal (5TP on AC/DC Gate PWB) will be within 1 volt of 0 VDC. Adjust the Transfer and Tracking deadband, potentiometer 1P to obtain close to this value.

While the DC SVA is tracking the AC Regulator, the 90R6/CS should not have any control.

90R1 Setting (Sh. 05B):

Set the On-Line pickup adjuster 90R1 at 1.0 volts or per sheet A04 and lock the knob. This will allow a nominal 2% rise in terminal volts when the line breaker is closed.

90P Setting (Sh.05C):

Set 90P by energizing 94Z-1. With the AC regulator in control, adjust 90P (slowly) until rated (or operating) terminal voltage is held. The output of the AC SVA will track the setting of 90P.

CHAPTER 4
ON-LINE CHECKS

The on-line tests and adjustments must be carried out when the generator line breaker is closed. The generator and turbine must be capable of carrying full load. Preliminary adjustments can be made with a low load, but a complete adjustment requires full load. If full load cannot be obtained at the present time, do not neglect these adjustments when full load can be obtained. This is very important for the proper operation of the regulator. These tests will require approximately 8 hours of time. Use a DVM as a transfer voltmeter at 5TP on the AC/DC PWB if one is not supplied.

____1. Ensure that "Synchronizing" checks are completed.

____2. Remove the URAL and Impedance Compensator PWB from the slots so that the pins are not making contact.

____3. The AC regulator must be in control when going on line. If the off line adjustments have been completed and a restart is made, the AC regulator will be in control when 52G closes. If the off line adjustments have been completed and the unit is to be synchronized to the line, make sure that 83SR-1 is energized.

____4. Match the generator voltage with the bus voltage before synchronizing. When the line breaker closes, observe the terminal voltage and VARS. If the pickup of voltage or VARS is too great, adjust 90R1.

****
* CAUTION *
****

90R1 IS CALIBRATED TO INCREASE THE GENERATOR TERMINAL VOLTAGE BY 2% PER 1 VOLT CHANGE. EXCESSIVE ADJUSTMENT OF 90R1 MAY FORCE THE GENERATOR TO CEILING

Do not adjust 90P unless the customer is always operating the generator at a voltage other than rated. Potentiometer 90P will control the AC Regular setpoint until 95% speed is reached. Once 94Z-1 is de-energized, the 90R1 and 90RR (90R4/CS) will be in control.

____5. The first time that the unit is synchronized, watch the generator line volts and field volts. If instability occurs, trip 52G.

Adjust 90R6 (and/or DC SVA) with 52GX-1A energized to set field volts to the approximate value at which instability occurred. Reclose 52G and if the problem reappears, transfer to the DC regulator to see if the problem continues.

_____6. Raise the MW output of the turbine with the governor and the voltage or VAR output of the generator with the AC SVA. Note: the AC (or DC) regulator does not effect the MW output of the turbine, but the turbine output may effect the VARs (and/or voltage) from the generator. Watch for oscillations in field volts.

The AC regulator should control the voltage or VARs smoothly with no oscillations or transient effects from light load up to peak load. Exercise the Raise/Lower controls to check the full range of operations.

****
* CAUTION *
****

DO NOT EXCEED THE GENERATOR CAPACITY DURING THIS EXERCISE. ALSO OPERATION IN THE UNDEREXCITED REGION MAY CAUSE A LOSS OF SYNCHRONISM.

_____7. Zero the transfer voltmeter with 90R6 (or DC SVA) and transfer to DC regulator. For DC SVA option, exercise controls to check the full range of operation. If the highest field voltage cannot be reached (Sh. A04-E2B) with the DC SVA, adjust SPAN potentiometer 2P on the AC/DC Gate PWB until the maximum can be achieved (CW to increase).

_____8. Adjust the output of the turbine/generator so that at least 50% of rated line current is carried. With the unit in DC regulator control and the Impedance Compensator on the extender, perform the Impedance Compensator checks (Page 4-3 through 4-8). Note: The generator load, MVARS, MW and I line must be constant for this calibration or the results may not agree with calculated values.

_____9. With the unit in DC regulator control, place the URAL PWB on the extender and perform the URAL checks (Page 4-9 to 4-11).

_____10. Perform the Auto Tracking checks (Page 4-11).

_____11. Perform the V/Hz protection checks (Page 4-11).

_____12. Perform the VAR PF controller checks (Page 4-12).

_____13. Adjust the output of the turbine-generator so that the field voltage is equal to the value on sheet A04-E2D. This procedure is best performed in AC regulator and when the point is reached, null the DC regulator with 90R6. (For units with DC SVA, set 90R6 to the output of the DC SVA and lock 90R6.) Record field volts and the -VF signal.

FLD Volts = _____ VDC -VF = _____VDC

_____14. Remove 83SR test switch circuit. Be careful not to cause unwanted transfers.

This completes all excitation checks.

Impedance Compensator Checks (Sh. 05A):

The following checks must be performed on the Impedance Compensator PWB before the unit is turned over to the customer.

The excitation system must be in the DC regulator mode and carrying overexcited VARS (pf<1.0, lagging).

For 1 Phase Sensing (Sh. 05A):

Connect the oscilloscope, CH1 to pins 11 and 12 on the Impedance Compensator PWB. Connect CH2 to 6TP. Use line synch. on scope. Adjust CH1 so that 1 cycle is 9 divisions.

For the correct operation CH2 (6TP, current) will lag CH1 (pins 11 and 12 voltage) by (90° Pf angle). The following table shows examples:

PF	ANGLE	CH2 LAGS CH1 (Degrees)
1.0	0	90
.94 (lag)	-20	110
.866(lag)	-30	120
.94 (lead)	+20	70

Calculate the power factor angle.

$$PF = COS \left[Tan^{-1} \frac{MVARS}{MW} \right]$$

If the above condition is not met, check the CT's, PT's and signal conditioner. If it is necessary to reverse a signal, use the CT signal and reverse it at SC 13 and SC 14. Remember URAL may be affected. The LCD/Droop jumper should be removed to prevent the X potentiometer from affecting output.

Place CH2 on 8TP. Now the CH2 signal will be 180° (inverted) from the CH1 signal. If this is not true, the board may be bad.

Remove the PT input to the Impedance Comp. by disconnecting the voltage signal at the Signal Conditioner SC 17.

Set the X potentiometer to zero. Set the R potentiometer to 10 (full CW).

Obtain the present generator phase 3 line current and calculate the voltage at 5TP.

$$V_{5TP} = \frac{I\ actual}{I\ rated} \times .5\ \frac{VDC}{base}$$

V_{5TP} = _____ X .5 = _____

Measure the voltage at 5TP and adjust it to the value calculated by turning potentiometer 3P. This is voltage V_{5R}.

V_{5R} = _____

Return the R potentiometer to 0.

Reconnect the PT signal at SC17.

Measure the voltage at 9TP. Adjust the voltage at 9TP to -5.0 VDC, with rated terminal voltage, using potentiometer 7P. If the terminal voltage is greater than rated, 9TP should be greater by the some percentage.

Calculate the voltage at 5TP from the following formula.

$$V_{5TP} = \sqrt{(V5R \sin \theta)^2 + (|V9TP| - V5R \cos \theta)^2}$$

Where θ = pf angle (neg. for lagging pf)

V5R = voltage measured during current calibration

$|V9TP|$ = voltage magnitude measured during voltage calibration

V5TP = _____(calculated)

Set the R potentiometer to 10.

Measure the voltage at 5TP. If it is not the same as calculated, recheck current and voltage calibrations.

V5TP = _____(measured)

Calculate the voltage at 2TP.

V2TP = -4 (V5TP + V9TP)

V2TP = _____

Measure the voltage at 2TP. If it is not the same as the calculated value, the 8A amplifier is bad.

With the regulator still in DC mode, take the Impedance Compensator off the extender, remove the extender and replace the Impedance Compensator in its proper slot.

With the generator carrying overexcited current, and the DC regulator in control, turning the R potentiometer clockwise will cause the transfer voltmeter to deflect in a direction to lower excitation. The voltage at 2TP (AC) on the AC/DC PWB will go more positive. The output at 2TP (OUT) on the Impedance Compensator will go negative.

Return the control to the AC regulator. With the generator carrying overexcited current, turning the R potentiometer clockwise will cause the excitation and the VARS (or voltage) to decrease.

Final adjustment of the compensator must be made on the basis of experience. Final adjustment can be made only after considerable experience with the generator operating under the control of the regulator. It is desirable to keep the amount of compensation to the minimum required for proper division of VAR's between generators to avoid excessive voltage regulation. Remember that the X potentiometer must be left on 0 when used in this manner.

The Impedance Compensator is used to compensate for the lack of a reactance between the generator and the system, or other generators. If a transformer (2 or 3 windings) is between each generator and the bus, the Impedance Compensator may not be needed (or the settings may be very low).

For 3 Phase Sensing (Sh. 05A):

Connect the oscilloscope, CH1 to pins 11 and 12 on the Impedance Compensator PWB. Connect CH2 to 6TP. Use Line Synch. on scope. Adjust CH1 so that 1 cycle is 9 divisions.

For the correct operation, CH2 (6TP, current) will lag CH1 (pins 11 and 12 voltage) by the power factor angle. The following table shows examples:

PF	ANGLE	CH2 LAGS CH1 (Degrees)
1.0	0	0
.94	-20	20
.866 (lag)	-30	30
.94 (lead)	+20	Leads by 20

Calculate the power factor angle.

$$pf = \cos\left[\tan^{-1}\frac{MVARS}{MW}\right]$$

If the above condition is not met, check the CT's, PT's and signal conditioner for proper phasing. If it is necessary to reverse a signal, use the CT signal at SC13 - SC14. The URAL checks may be affected by signal reversal.

Place CH1 on 8TP. Now the CH2 signal (6TP, current) will lead CH1 by (180 - pf angle). If this is not true, check the previous steps, or the board may be bad.

Remove the PT signal input to the Impedance Compensator PWB by disconnecting the voltage signal at the Signal Conditioner SC17.

Obtain the present generator phase 3 line current and calculate the voltage V5R.

$$V5R = \frac{I\ Actual}{I\ Rated\ Base} \times .5\ VDC$$

V5R = _____ X .5 = _____

(This voltage is measured at 5TP)

Set the R potentiometer at 10. Measure the voltage at 5TP. Adjust the voltage at 5TP to the value calculated, V5R, with potentiometer 3P.

V5R = _____ (measured)

Return the R potentiometer to 0.

Set Jumper to LDC (it will be set in its final position later).

Place CH1 of the oscilloscope on 6TP. Adjust the time base until 1/2 cycle is 9 divisions.

Place CH2 on 3TP. The voltage at 3TP should lag CH1 by 45 degrees (2 1/4 divisions). Adjust this angle with potentiometer 4P.

Now place CH2 on 4TP. The voltage at 4TP should lead CH1 by 90 degrees (4 1/2 divisions). Adjust this angle with potentiometer 5P.

Calculate the voltage V5X.

$$V5X = \frac{I\ Actual}{I\ Rated\ Base} \times 1.25\ VDC$$

V5X = _____ X 1.25 = _____

(This voltage is measured at 5TP)

Set the X potentiometer at 25 and measure the voltage at 5TP. Adjust the voltage at 5TP to the value calculated, V5X, with potentiometer 6P. V5TP will be a positive voltage.

V5X = _____ (measured)

Return the X potentiometer to zero.

Reconnect the voltage signal at SC17.

Measure the voltage at 9TP. Adjust the voltage at 9TP to -5.0 VDC with rated terminal voltage on the generator, with potentiometer 7P. If the terminal voltage is not at rated, adjust 9TP by the same percentage.

Calculate the voltage at 5TP from the following formula.

$$V5TP = \sqrt{(V_{5X}\text{Cos}\theta)^2 + (|V_{9TP}| + V_{5X}\text{Sin}\theta)^2}$$

V5TP = _____ (Calculated)

Where θ is the measured PF angle and is negative when the machine is overexcited, V_{5X} is the measured voltage on 5TP during the previous test and V_{9TP} is the magnitude of the voltage on 9TP (this should be 5V if the generator is at rated voltage).

Set the X potentiometer to 25.

Measure the voltage at 5TP. If it is not the same as calculated, recheck current and voltage calibrations. It will be approximately +5 VDC.

V5TP = _____(Measured)

Calculate the voltage at 2TP.

$V_{2TP} = -4$ (V5TP + V9TP) NOTE: the signs are important.

V2TP = _____

Measure the voltage at 2TP. If it is not the same as calculated, amplifier 8A is bad.

Set the X potentiometer to 0 and the R potentiometer to 10.

Calculate the voltage at 5TP.

$$V_{5TP} = \sqrt{(V_{5R}\text{Sin}\theta)^2 + (|V_{9TP}| - V_{5R}\text{Cos}\theta)^2}$$

V_{5R} is the measured value of the voltage at 5TP recorded earlier when the voltage signal was zero and the R potentiometer was turned up.

V5TP = _____ (Calculated)

Measure the voltage at 5TP. If it is not the same as calculated, recheck current and voltage calibrations. It will be approximately +5.0 VDC.

V_{5TP} = _____(Measured).

At this point it must be determined whether the customer wants Reactive Current Compensation (RCC) or Active and Reactive Current Compensation (ARCC). RCC or Droop is used to parallel AC generators onto the same bus with little or no impedance between the tie point and the generator. RCC will simulate this reactance so that the generators can share reactive current. ARCC or Line Drop Compensation (LDC) is used to compensate for excessive reactance between the generator and the system tie point. It will provide compensation for real (I^2R) drops and reactive drops. Once the selection has been made (use Droop if in doubt) place the jumper on the Impedance Compensator PWB in the correct position.

For RCC (Set jumper to Droop):

With the generator carrying overexcited current, and the DC regulator in control, turning the X potentiometer clockwise will cause the transfer voltmeter to deflect in a direction to lower excitation. The voltage at 2TP (AC) on the AC/DC PWB will go more positive. The output at 2TP (OUT) on the Impedance Compensator will go negative.

Replace the Impedance Compensator PWB directly in its slot. Set knobs at 0. Return the control to the AC regulator. With the generator carrying overexcited current, turning the X potentiometer clockwise will cause the excitation and the VARS (or voltage) to decrease.

Final adjustment can be made only after considerable experience with the generator operating under the control of the regulator. It is desirable to keep the amount of compensation to the minimum required for proper division of VAR's between generators to avoid excessive voltage regulation. Remember that the R potentiometer must be left on 0 when used in this manner.

For ARCC (Set jumper on LDC):

With the generator carrying overexcited current, and the DC regulator in control, turning the X potentiometer clockwise will cause the transfer voltmeter to deflect in a direction to raise excitation. The voltage at 2TP (AC) on the AC/DC PWB will go more negative. The output at 2TP (OUT) on the Impedance Compensator will go positive.

If the generator is at unity power factor, or carrying only real power, turning the R potentiometer clockwise will result in the same condition as above.

Replace the Impedance Compensator PWB directly in its slot. Set knobs to 0. Return the control to the AC regulator. With the generator carrying overexcited current, turning the R or X potentiometer clockwise will cause the excitation and the VARS (or voltage) to increase.

Final adjustment of the compensator must be made on the basis of experience. If used for line drop compensation (LDC) preliminary adjustment can be made in accordance with the known values of resistance and reactance for that portion of the system over which compensation is desired. The R and X potentiometer faceplates are marked in percent, if the foregoing tests have been made correctly. Typically, line resistance is so low that the R potentiometer is set at zero. If the transformer reactance is 0.12 PU and it is desired to have about 0.06 PU reactance for proper VAR sharing, set the X potentiometer at 6.

URAL Adjustments (Sh. 05B):

From pre-excitation checks, the URAL calculations are as follows:

B = _____

R = _____

C = _____

1TB = _____ to _____

The following adjustments must be performed on the URAL PWB before the unit is turned over to the customer.

The excitation system must be in the DC regulator while these checks are being performed. The generator output must be stable while these measurements are made.

Obtain the present generator phase 3 line current, and calculate the voltage at 5TP.

$$V5TP = \frac{I\ Actual}{I\ Rated} \times 2.5$$

Base

V5TP = _____

Note: Ensure that the generator is at rated voltage and that the voltage at 8TP is 5.66 VRMS. Adjust with VCAL.

Set potentiometer 1P, CENTER, full CW. Ensure that the wiper is adjusted for zero volts.

Measure the voltage at 5TP. Set this voltage to the calculated value by potentiometer 4P, ICAL.

Check the phasing of the input signals by the following procedure:

For 1 Phase Sensing

At unity power factor (pf = 1), the voltage at pins 15 and 16 will lag the voltage at 6TP (Iin) by 90°. If the power factor is lagging, the voltage will lag by (90° - power factor angle). The voltage at pins 13 and 14 will lead the voltage at 6TP (Iin) by 90°. If the power factor is lagging, the voltage will lead by (90° + power factor angle).

For 3 Phase Sensing

The voltages should reach a peak in this order (0V com for ref). Pins 15 and 16 (Phase 1), 13, 14 (Phase 2) and 11,12 (Phase 3). Each should be 17 VRMS at rated generator volts.

Place CH1 of the oscilloscope on 9TP (Phase 3). Place CH2 on 6TP (Iin). When the power factor is lagging, the current (CH2) will lag the voltage (CH1). If corrections to the signal conditioner are necessary, remember that they may affect the Impedance Compensator. Check the angle B by measuring the lag between 9TP and 8TP.

Set the circle radius R. First measure the PT secondary voltage. Calculate the ratio between rated and actual. This ratio is $\triangle V$ = _____ (It is greater than 1.0 when the PT volts is greater than rated).

Calculate the voltage at 4TP.

$V4TP = 2.5 \times R \times \triangle V = 2.5 \times$ _____ $\times$ _____ = _____

Measure the voltage at 4TP and adjust to the calculated value using potentiometer 3P, RADIUS.

Set the circle center, C. Calculate the voltage at 5TP.

$V5TP = 2.5 \times C \times \triangle V$

V5TP = _____

Short the CT or remove the CT signal at SC13. Measure the voltage at 5TP and adjust to the calculated value using potentiometer 1P, CENTER.

Replace the CT signal.

To check these settings, continue to control the generator voltage with the DC Regulator. Carefully decrease excitation until the 3TP on the URAL PWB goes in the positive direction. This shows that the URAL circuit would take over if the AC Regulator were in control. The voltage at 5TP will be greater than the voltage at 4TP.

Replace the URAL PWB in the rack.

Set the URAL gain potentiometer 5P fully counterclockwise.

Stability of the URAL must now be checked. With the generator loaded so the URAL would not be in control, null the Transfer Voltmeter, using 90R4/CS and transfer to AC Regulator.

Gradually decrease excitation voltage until the further decreases do not effect the voltage. It should not be possible to reduce VAR's any further. Check this point against the characteristic curve previously calculated. There should be no oscillations present. The field voltage and Transfer Voltmeter on the GCP will show these if they are there. These meters, being analog, are better for this test. If oscillations are present, back off the limit if possible. If oscillations are too severe, immediately go back to DC Regulator and increase excitation to a safe value. Reduce the gain of the URAL, using potentiometer 5P, marked GAIN, and try again. The AC Regulator gain may be reduced to K3 = .12 to reduce oscillations.

Repeat this test for various values of active current and record for future reference.

If the curve actually obtained is not sufficiently close to that calculated, move the radius and center adjustments to achieve the desired characteristics.

Auto Tracking Checks (Sh. 06C):

Verify the proper operation of the Auto Tracking function. As MW load and/or VAR load is raised, the field voltage will increase (in the AC Regulator). The Transfer Voltmeter will deflect to the right. After 30 seconds (from the initial point of operation) the Transfer Voltmeter will begin to move to the left. It should stop close to zero. During the time that the AC Regulator is in control, the 90R6/CS will not affect the DC Regulator or Transfer Voltmeter.

When the Transfer Voltmeter is close to zero, a transfer to the DC Regulator can be made. If the AC/DC PWB and the Transfer and Tracking PWB are set up correctly, no appreciable change in field volts or VARS should occur (the field voltmeter needle may jump slightly) now 90R6/CS controls the excitation.

The light controlled by 83RM should designate the control switch that the operator should use.

If all the checks above are satisfactory, no further adjustment of the Auto Tracking function are necessary.

V/Hz Protection Checks (Sh. 10C):

The Volts/Hertz protection circuits can be exercised while on line to check their functions by closing the Test switch. This will add a voltage to the output of the V/Hz detector, simulating a high V/Hz condition. No trip will occur while the Test switch is closed.

1. Close the test switch and measure the voltage at 2TP.

2. With the TEST potentiometer 1P, adjust this voltage to -5.80 VDC.

3. Open test switch and reset.

4. Close the test switch and start timing.

5. 60 seconds later the Trip LED should flash. During this time relay 59VH-1X should be energized.

6. Using the test potentiometer, 1P, adjust 2TP to -5.95 VDC and open the Test switch and reset.

7. Close the test switch again.

8. The trip light should flash 2 seconds later.

9. If the above times are not correct, recalibrate the V/Hz trip boards by the pre-excitation checks, except use the test potentiometer as the variable input.

VAR-PF Controller Checks (Sh. 05G):

The VAR-PF Controller should be calibrated and adjusted per GEK-36432.

To permit the VAR-PF controller to control the AC SVA, relay 43VAX must be energized and the unit must be on line (52GX-1A energized).

Perform the calibrations and adjustments per GEK-36432.

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